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Batch: B2 Data Science

Section 1: Visualization over Built-in Datasets

Basic visual analysis using a single script `section1.py` generating Histograms, KDE plots, Scatter plots, and Bar plots for all three datasets.

Code Implementation (`section1.py`)

Code:

```
python

import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import os

os.makedirs('images', exist_ok=True)

# --- Tips Dataset Plots ---
print("--- PLOTTING TIPS ---")
tips_df = sns.load_dataset('tips')
print(tips_df.head(3))

plt.figure()
sns.histplot(tips_df['total_bill'], kde=False, color='coral')
plt.title('Total Bill Distribution')
plt.savefig('images/tips_section1_hist.png')
plt.close()

plt.figure()
sns.kdeplot(tips_df['tip'], fill=True, color='mediumorchid')
plt.title('Tip Density Plot')
plt.savefig('images/tips_section1_kde.png')
plt.close()

plt.figure()
sns.scatterplot(data=tips_df, x='total_bill', y='tip', hue='smoker')
plt.title('Total Bill vs Tip')
```

```

plt.savefig('images/tips_section1_scatter.png')
plt.close()

plt.figure()
sns.barplot(data=tips_df, x='day', y='total_bill', hue='day', legend=False)
plt.title('Average Bill by Day')
plt.savefig('images/tips_section1_bar.png')
plt.close()

# --- Flights Dataset Plots ---
print("\n--- PLOTTING FLIGHTS ---")
flights_df = pd.read_csv('../flights.csv')
print(flights_df.head(3))

flights_df['dep_delay'] =
flights_df['dep_delay'].fillna(flights_df['dep_delay'].mean())
flights_df['arr_delay'] =
flights_df['arr_delay'].fillna(flights_df['arr_delay'].mean())

plt.figure()
sns.histplot(flights_df['dep_delay'], kde=False, color='seagreen', bins=50)
plt.title('Departure Delay Histogram')
plt.savefig('images/flights_section1_hist.png')
plt.close()

plt.figure()
sns.kdeplot(flights_df['arr_delay'], fill=True, color='indianred')
plt.title('Arrival Delay KDE')
plt.savefig('images/flights_section1_kde.png')
plt.close()

plt.figure()
sns.scatterplot(data=flights_df, x='dep_delay', y='arr_delay', hue='month',
palette='tab10', alpha=0.6)
plt.title('Departure Delay vs Arrival Delay')
plt.savefig('images/flights_section1_scatter.png')
plt.close()

plt.figure()
sns.barplot(data=flights_df, x='month', y='dep_delay', hue='month', legend=False)
plt.title('Avg Departure Delay by Month')
plt.savefig('images/flights_section1_bar.png')

```

```

plt.close()

# --- Iris Dataset Plots ---
print("\n--- PLOTTING IRIS ---")
iris_df = sns.load_dataset('iris')
print(iris_df.head(3))

plt.figure()
sns.histplot(iris_df['sepal_length'], kde=False, color='royalblue')
plt.title('Sepal Length Frequency')
plt.savefig('images/section1_hist.png')
plt.close()

plt.figure()
sns.kdeplot(iris_df['sepal_width'], fill=True, color='darkorange')
plt.title('Sepal Width KDE')
plt.savefig('images/section1_kde.png')
plt.close()

plt.figure()
sns.scatterplot(data=iris_df, x='sepal_length', y='sepal_width', hue='species')
plt.title('Sepal Length vs Sepal Width')
plt.savefig('images/section1_scatter.png')
plt.close()

plt.figure()
sns.barplot(data=iris_df, x='species', y='petal_length', hue='species',
            legend=False)
plt.title('Avg Petal Length by Species')
plt.savefig('images/section1_bar.png')
plt.close()

```

Output:

```

text
--- PLOTTING TIPS ---
   total_bill  tip  sex smoker  day    time  size
0      16.99  1.01 Female    No  Sun  Dinner    2
1      10.34  1.66  Male    No  Sun  Dinner    3
2      21.01  3.50  Male    No  Sun  Dinner    3

--- PLOTTING FLIGHTS ---

```

```

    id year month day ... dep_delay arr_time sched_arr_time arr_delay
0    1  2013     1   1 ...         2.0    830.0           819        11.0
1    2  2013     1   1 ...         4.0    850.0           830        20.0
2    3  2013     1   1 ...         2.0    923.0           850        33.0

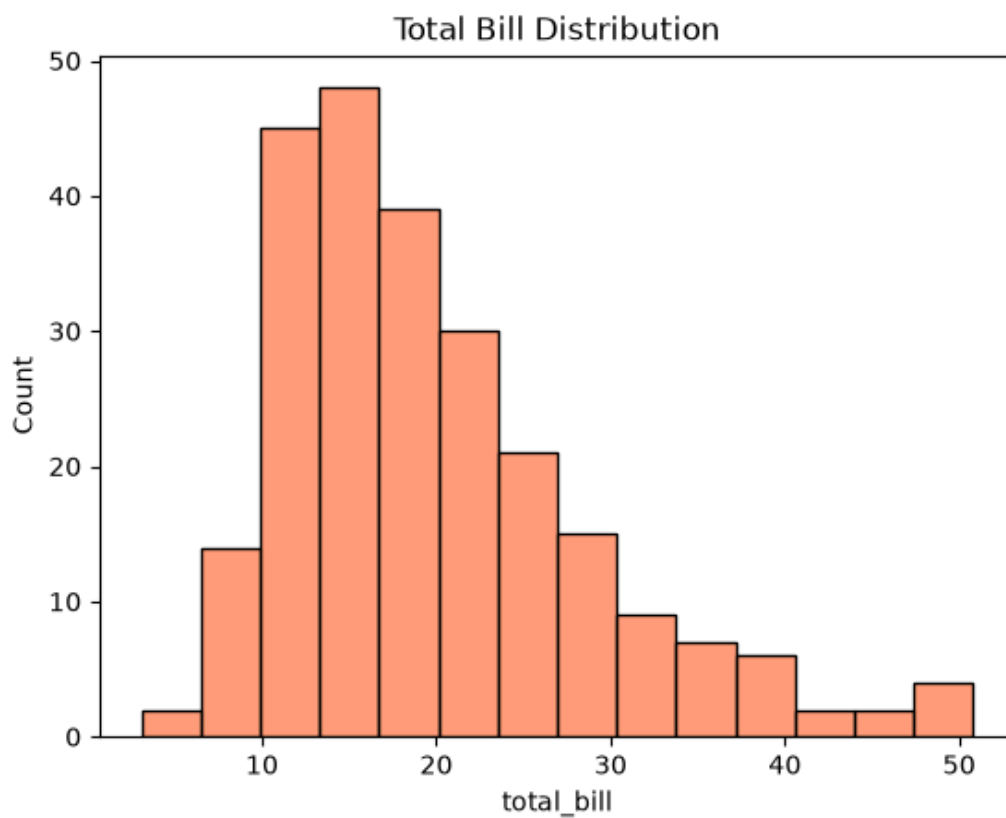
[3 rows x 10 columns]

--- PLOTTING IRIS ---
    sepal_length sepal_width petal_length petal_width species
0             5.1         3.5         1.4         0.2  setosa
1             4.9         3.0         1.4         0.2  setosa
2             4.7         3.2         1.3         0.2  setosa

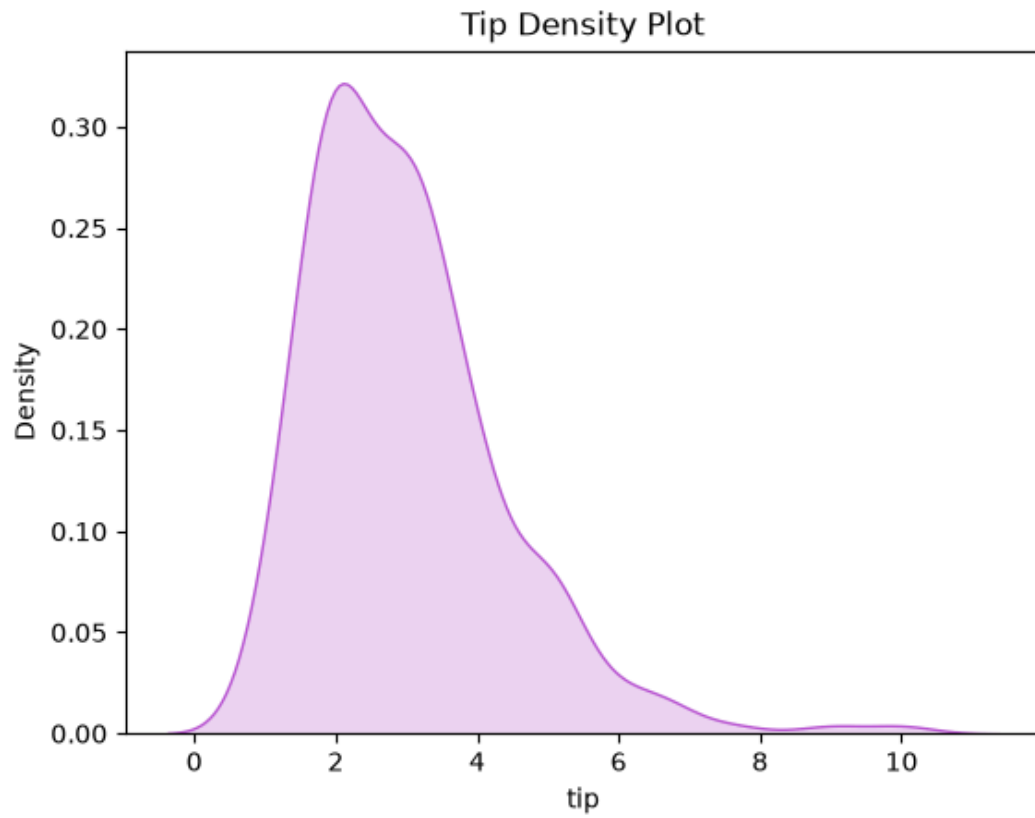
```

Visuals (Tips):

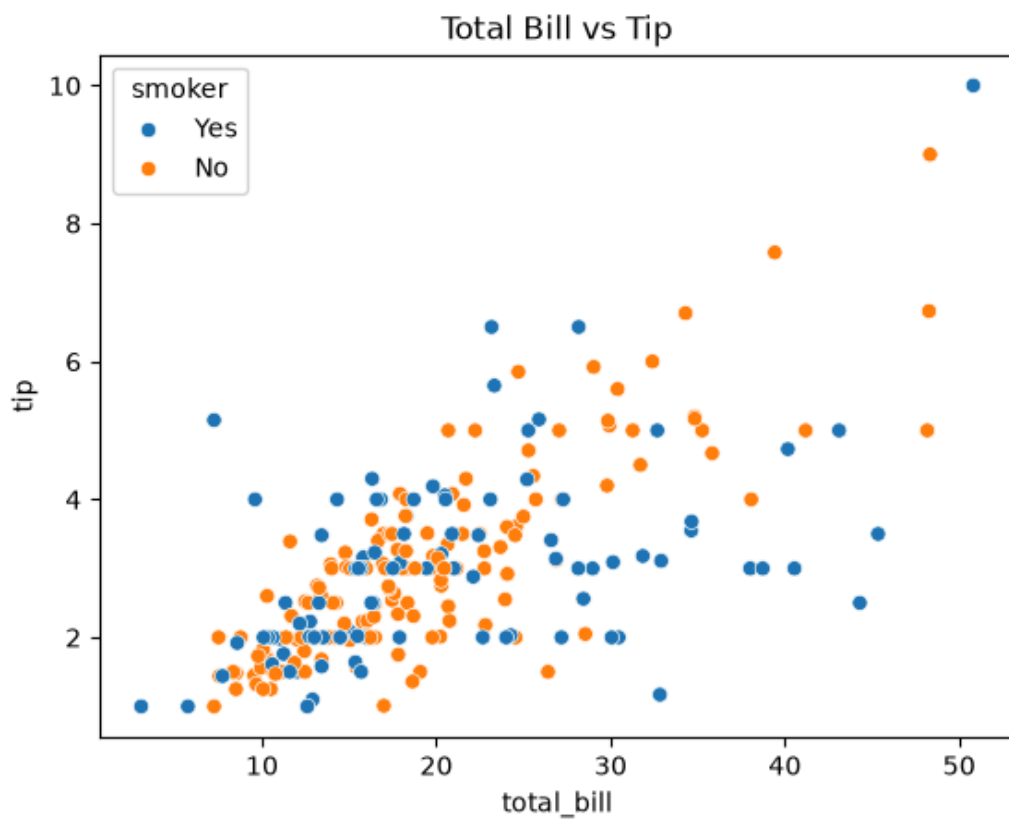
Tips Total Bill Histogram



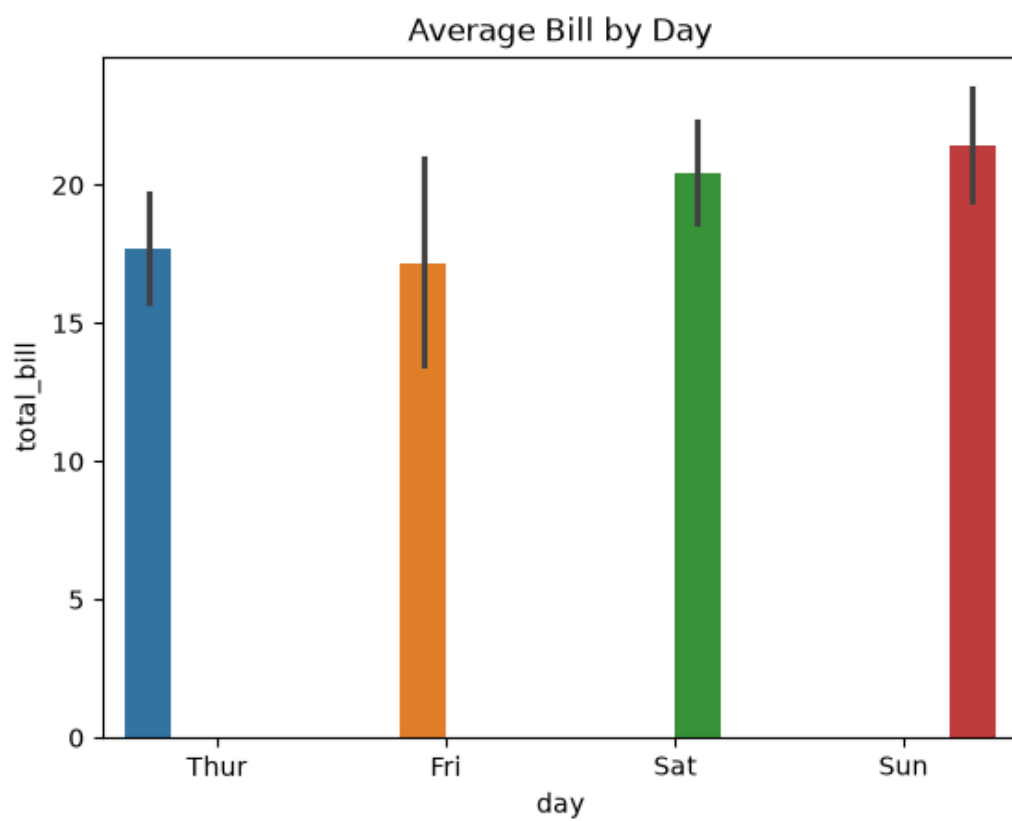
Tips Amount KDE Plot



Tips Total Bill vs Tip Scatter Plot

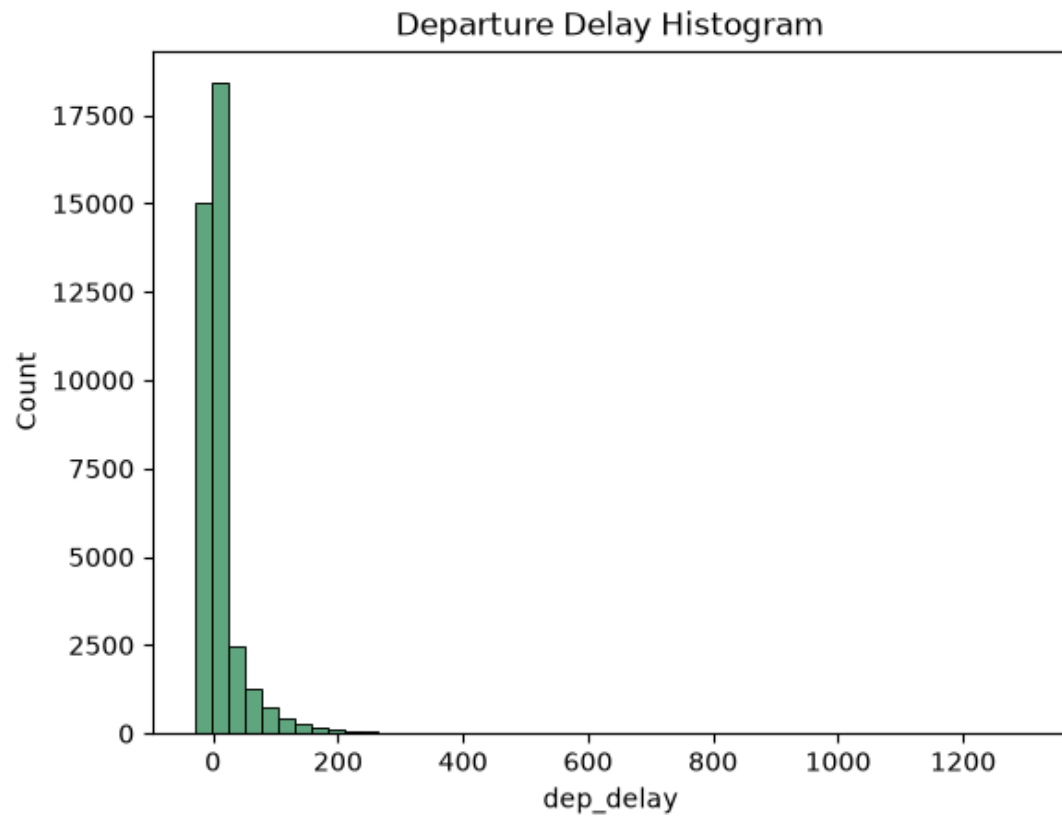


Tips Average Bill by Day Bar Plot

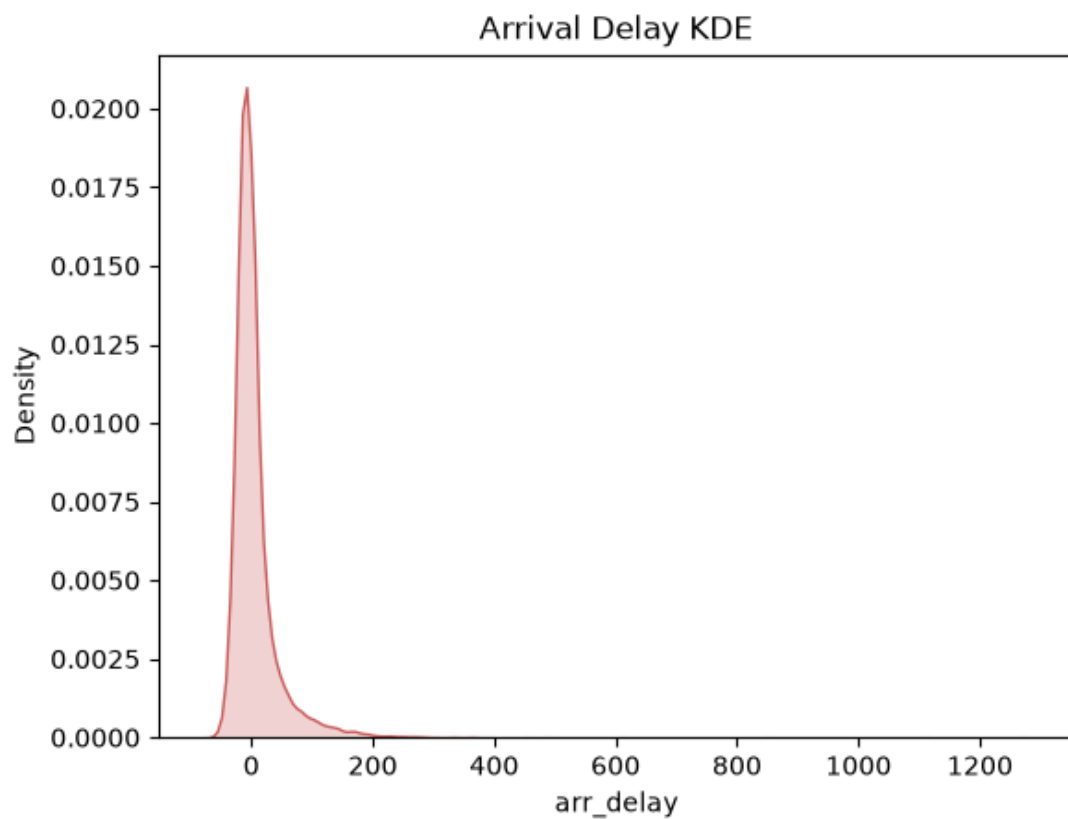


Visuals (Flights):

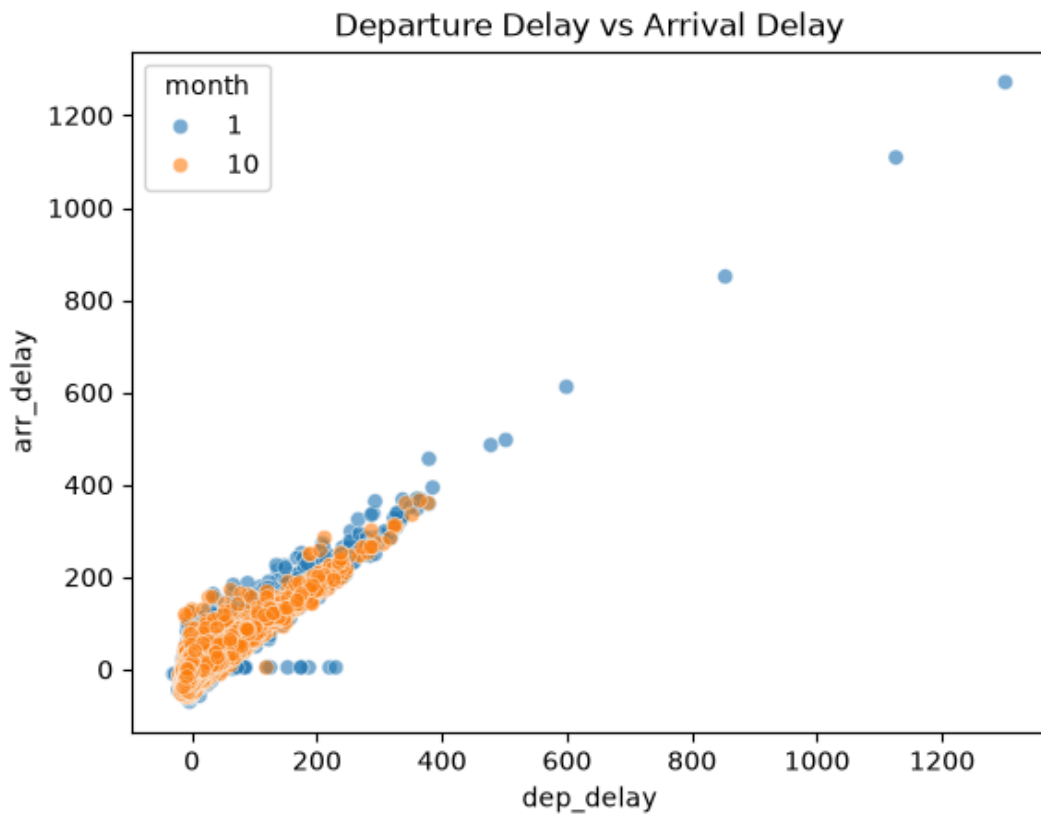
Flights Departure Delay Histogram



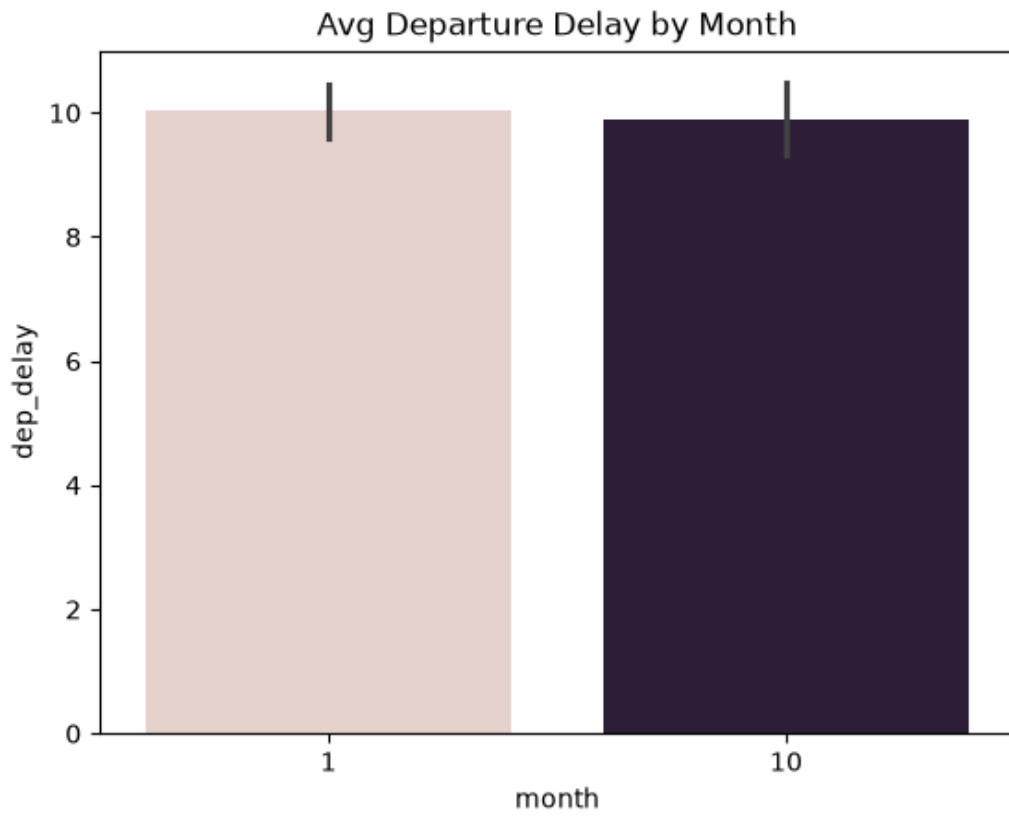
Flights Arrival Delay KDE Plot



Flights Scatter Plot

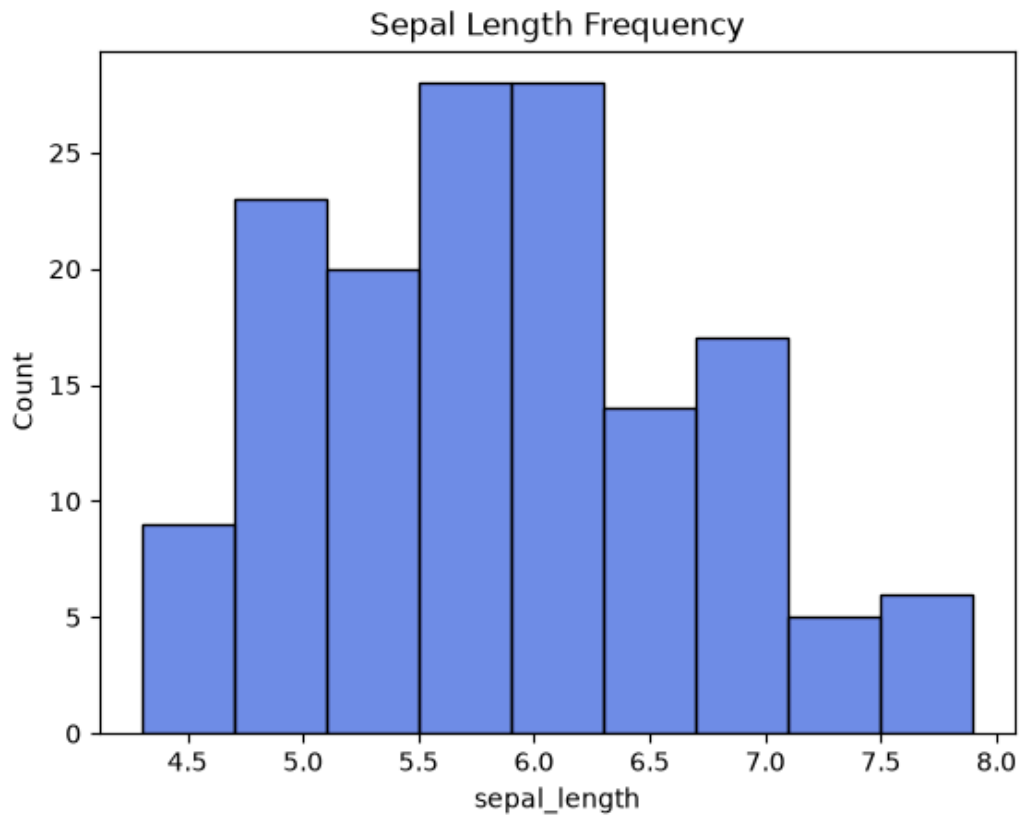


Flights Average Departure Delay by Month Bar Plot

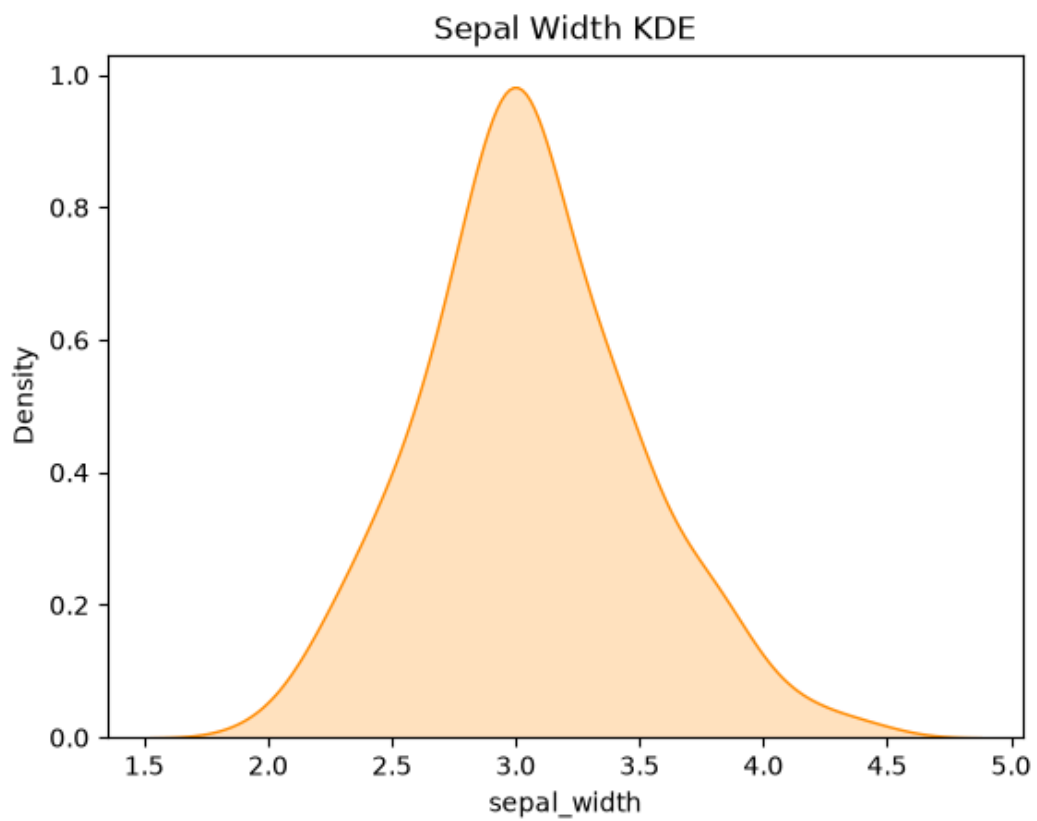


Visuals (Iris):

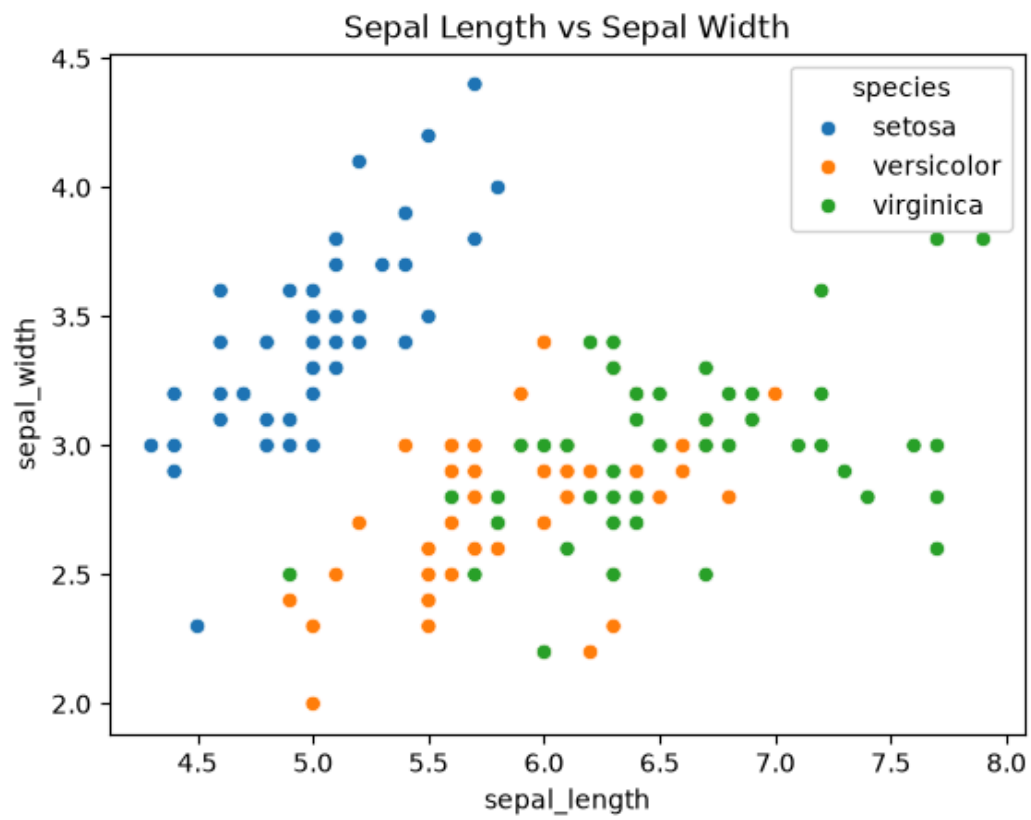
Iris Sepal Length Histogram



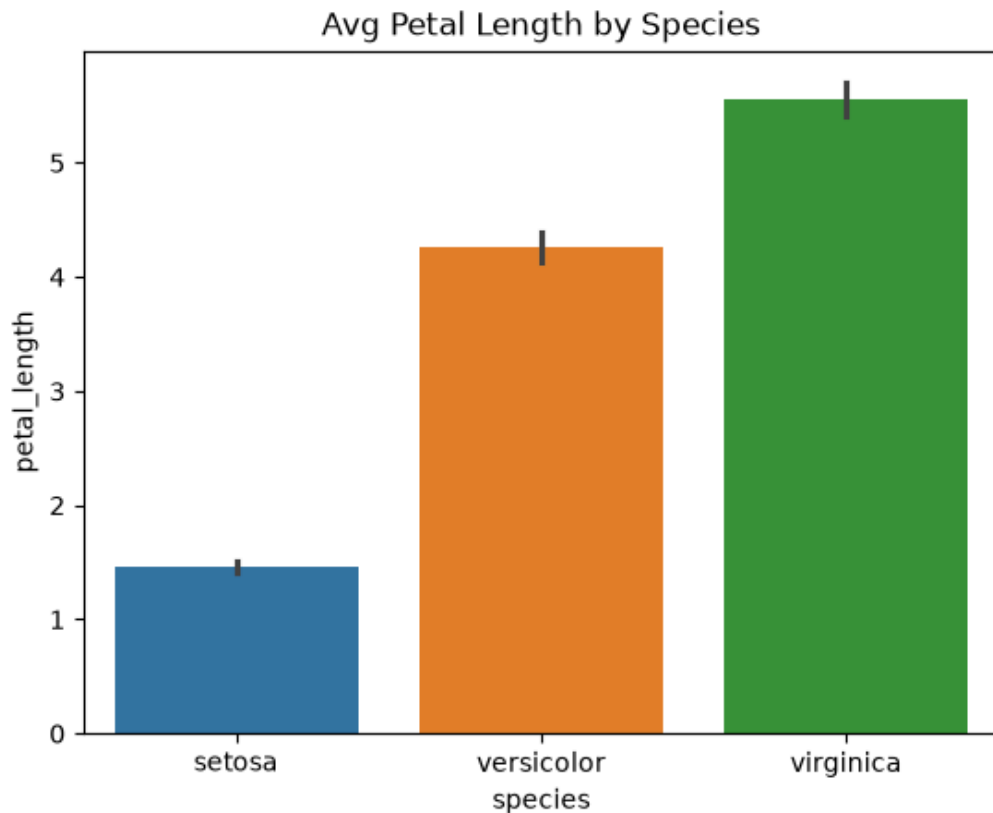
Iris Sepal Width KDE Plot



Iris Scatter Plot



Iris Average Petal Length by Species Bar Plot



Section 2: Practice with Different Plot Categories

This section plots distributions, categorical indicators, heatmaps, clustermaps, and regressions using `section2.py` for all three datasets.

Code Implementation (`section2.py`)

Code:

```
python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import os

os.makedirs('images', exist_ok=True)

# --- Category & Regression Plots for Tips ---
tips_df = sns.load_dataset('tips')

plt.figure()
```

```
sns.histplot(data=tips_df, x='total_bill', kde=True)
plt.savefig('images/tips_section2_dist_hist.png')
plt.close()

plt.figure()
sns.kdeplot(data=tips_df, x='tip', hue='sex', fill=True)
plt.savefig('images/tips_section2_dist_kde.png')
plt.close()

plt.figure()
sns.barplot(data=tips_df, x='day', y='total_bill', hue='day', legend=False)
plt.savefig('images/tips_section2_cat_bar.png')
plt.close()

plt.figure()
sns.countplot(data=tips_df, x='day', hue='day', legend=False)
plt.savefig('images/tips_section2_cat_count.png')
plt.close()

plt.figure()
sns.boxplot(data=tips_df, x='day', y='total_bill', hue='day', legend=False)
plt.savefig('images/tips_section2_cat_box.png')
plt.close()

plt.figure()
sns.violinplot(data=tips_df, x='day', y='total_bill', hue='day', legend=False)
plt.savefig('images/tips_section2_cat_violin.png')
plt.close()

plt.figure()
sns.swarmplot(data=tips_df, x='day', y='total_bill', size=4.5, hue='day',
legend=False)
plt.savefig('images/tips_section2_cat_swarm.png')
plt.close()

plt.figure()
corr_tips = tips_df.select_dtypes(include=['float64', 'int64']).corr()
sns.heatmap(corr_tips, annot=True, cmap='mako')
plt.savefig('images/tips_section2_matrix_heatmap.png')
plt.close()

plt.figure()
sns.regplot(data=tips_df, x='total_bill', y='tip', color='salmon')
```

```

plt.savefig('images/tips_section2_reg_plot.png')
plt.close()

sns.lmplot(data=tips_df, x='total_bill', y='tip', hue='smoker')
plt.savefig('images/tips_section2_reg_lmplot.png')
plt.close()

# --- Category & Matrix Plots for Flights (Heatmap & Clustermap) ---
flights_df = pd.read_csv('../flights.csv')
flights_df['dep_delay'] =
flights_df['dep_delay'].fillna(flights_df['dep_delay'].mean())
flights_df['arr_delay'] =
flights_df['arr_delay'].fillna(flights_df['arr_delay'].mean())

plt.figure()
sns.histplot(data=flights_df, x='dep_delay', kde=True, bins=50)
plt.savefig('images/flights_section2_dist_hist.png')
plt.close()

flights_subset = flights_df[flights_df['month'].isin([1, 2, 3])]
plt.figure()
sns.kdeplot(data=flights_subset, x='arr_delay', hue='month', fill=True,
palette='muted')
plt.savefig('images/flights_section2_dist_kde.png')
plt.close()

plt.figure()
sns.barplot(data=flights_df, x='month', y='dep_delay', hue='month', legend=False,
palette='coolwarm')
plt.savefig('images/flights_section2_cat_bar.png')
plt.close()

plt.figure()
sns.countplot(data=flights_df, x='month', hue='month', legend=False,
palette='magma')
plt.savefig('images/flights_section2_cat_count.png')
plt.close()

plt.figure()
sns.boxplot(data=flights_subset, x='month', y='dep_delay', hue='month',
legend=False)
plt.savefig('images/flights_section2_cat_box.png')

```

```

plt.close()

plt.figure()
sns.violinplot(data=flights_subset, x='month', y='arr_delay', hue='month',
legend=False)
plt.savefig('images/flights_section2_cat_violin.png')
plt.close()

flights_sample = flights_subset.sample(200, random_state=42)
plt.figure()
sns.swarmplot(data=flights_sample, x='month', y='dep_delay', hue='month',
legend=False, size=4.5)
plt.savefig('images/flights_section2_cat_swarm.png')
plt.close()

# Month vs Day pivot delay values
flights_pivot = flights_df.pivot_table(index='month', columns='day',
values='dep_delay', aggfunc='mean').fillna(0)

# Heatmap
plt.figure(figsize=(10, 6))
sns.heatmap(flights_pivot, cmap='coolwarm', fmt=".2f")
plt.tight_layout()
plt.savefig('images/flights_section2_matrix_heatmap.png')
plt.close()

# Clustermap
g = sns.clustermap(flights_pivot, cmap='coolwarm', figsize=(8, 8))
g.savefig('images/flights_section2_matrix_clustermap.png')
plt.close()

plt.figure()
sns.regplot(data=flights_sample, x='dep_delay', y='arr_delay', color='purple')
plt.savefig('images/flights_section2_reg_plot.png')
plt.close()

sns.lmplot(data=flights_sample, x='dep_delay', y='arr_delay', hue='month')
plt.savefig('images/flights_section2_reg_lmplot.png')
plt.close()

# --- Category Plots for Iris ---
iris_df = sns.load_dataset('iris')

```

```
plt.figure()
sns.histplot(data=iris_df, x='petal_length', kde=True)
plt.savefig('images/section2_dist_hist.png')
plt.close()

plt.figure()
sns.kdeplot(data=iris_df, x='petal_width', hue='species', fill=True)
plt.savefig('images/section2_dist_kde.png')
plt.close()

plt.figure()
sns.barplot(data=iris_df, x='species', y='sepal_length', hue='species',
legend=False)
plt.savefig('images/section2_cat_bar.png')
plt.close()

plt.figure()
sns.countplot(data=iris_df, x='species', hue='species', legend=False)
plt.savefig('images/section2_cat_count.png')
plt.close()

plt.figure()
sns.boxplot(data=iris_df, x='species', y='sepal_width', hue='species', legend=False)
plt.savefig('images/section2_cat_box.png')
plt.close()

plt.figure()
sns.violinplot(data=iris_df, x='species', y='petal_length', hue='species',
legend=False)
plt.savefig('images/section2_cat_violin.png')
plt.close()

plt.figure()
sns.swarmplot(data=iris_df, x='species', y='petal_width', size=4.5, hue='species',
legend=False)
plt.savefig('images/section2_cat_swarm.png')
plt.close()

plt.figure()
corr_iris = iris_df.iloc[:, :4].corr()
sns.heatmap(corr_iris, annot=True, cmap='rocket')
plt.savefig('images/section2_matrix_heatmap.png')
```

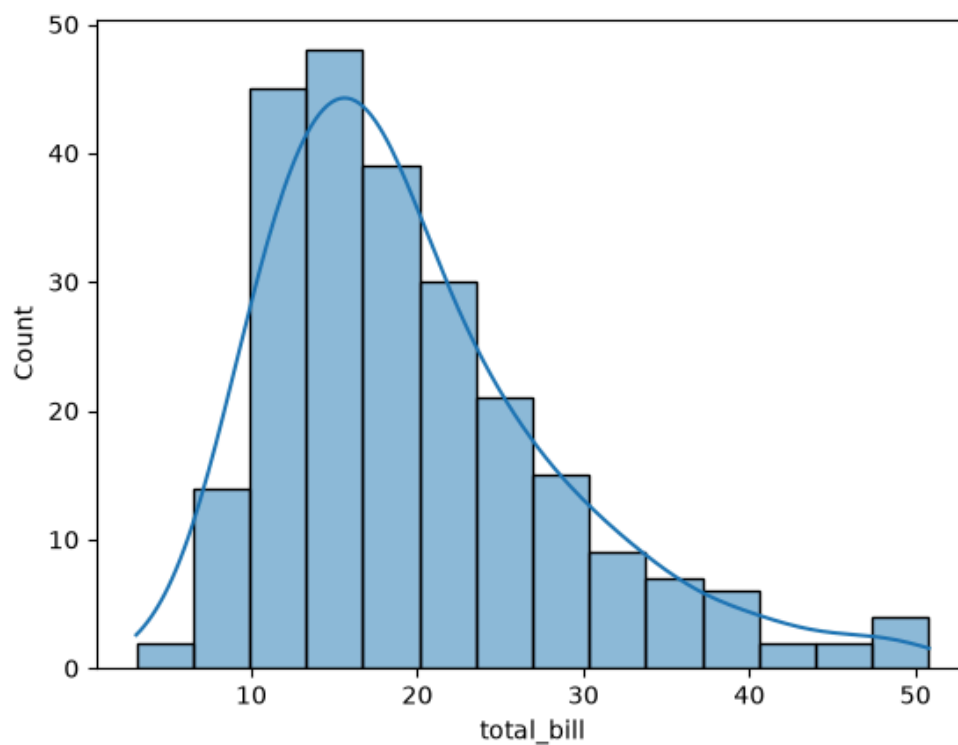
```
plt.close()

plt.figure()
sns.regplot(data=iris_df, x='sepal_length', y='petal_length', color='teal')
plt.savefig('images/section2_reg_plot.png')
plt.close()

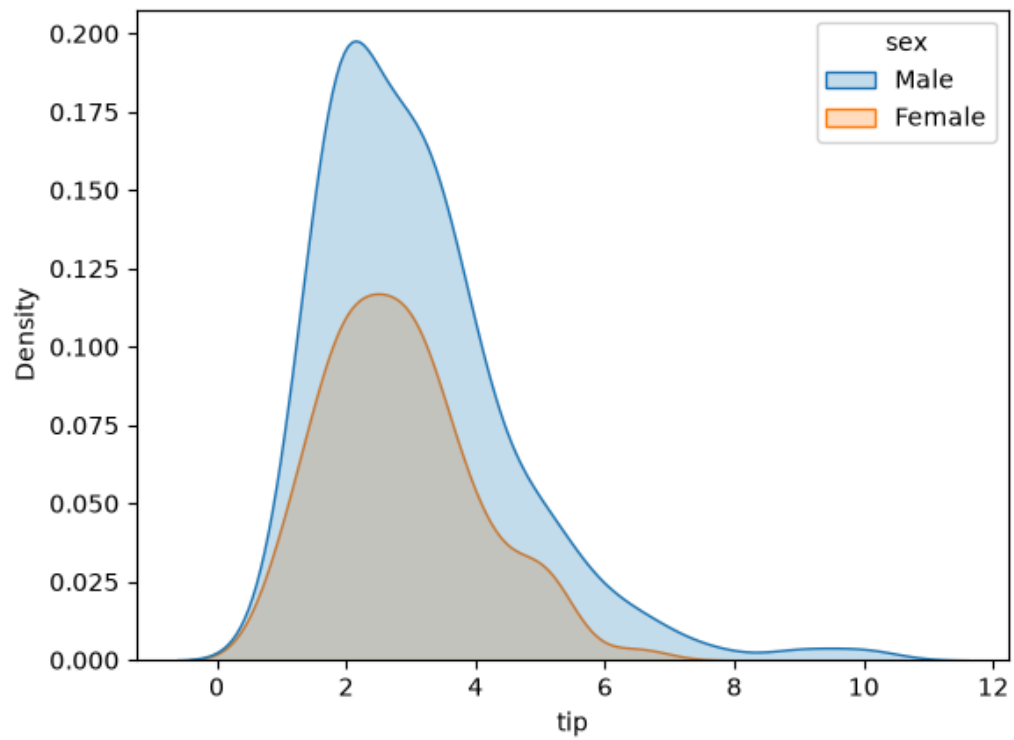
sns.lmplot(data=iris_df, x='sepal_length', y='petal_length', hue='species')
plt.savefig('images/section2_reg_lmplot.png')
plt.close()
```

Visuals (Tips):

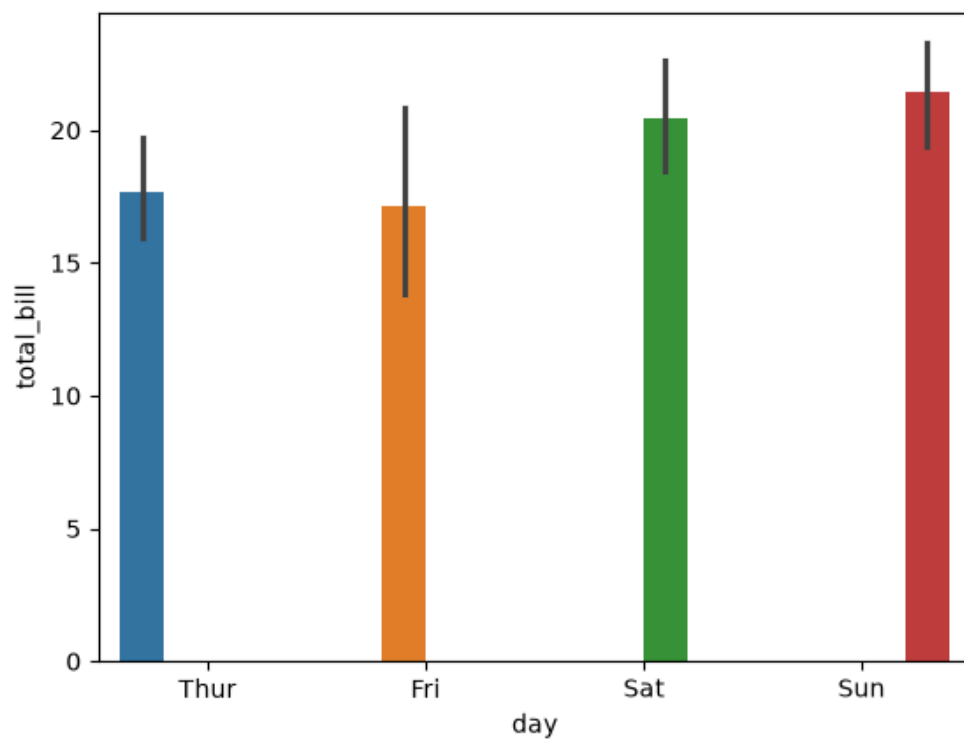
- Distribution Histplot:



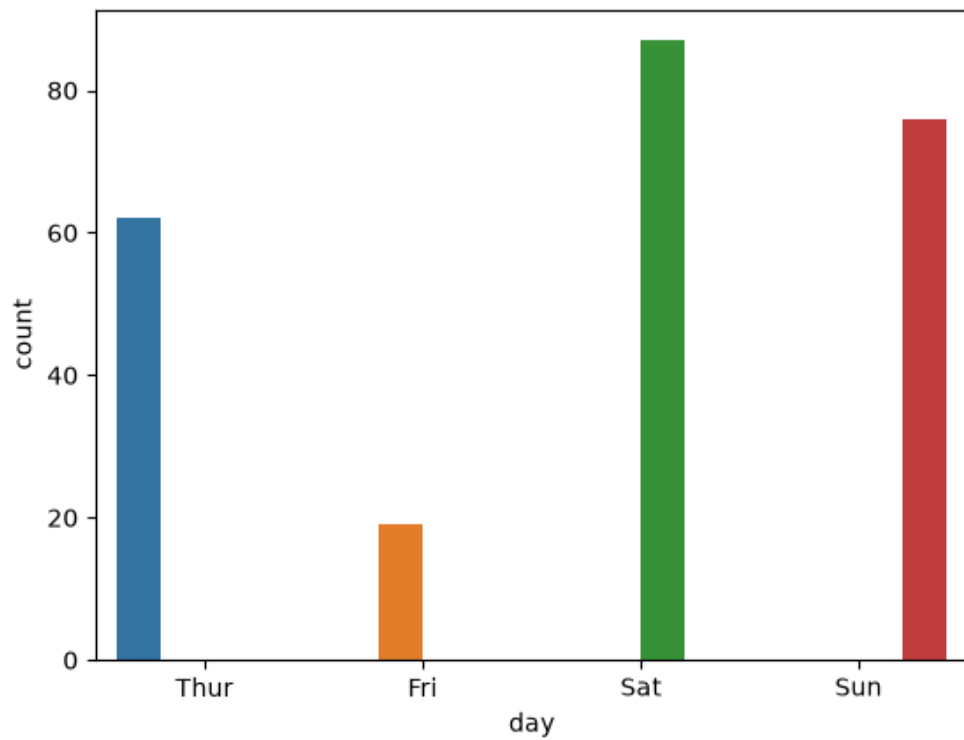
- Distribution KDE:



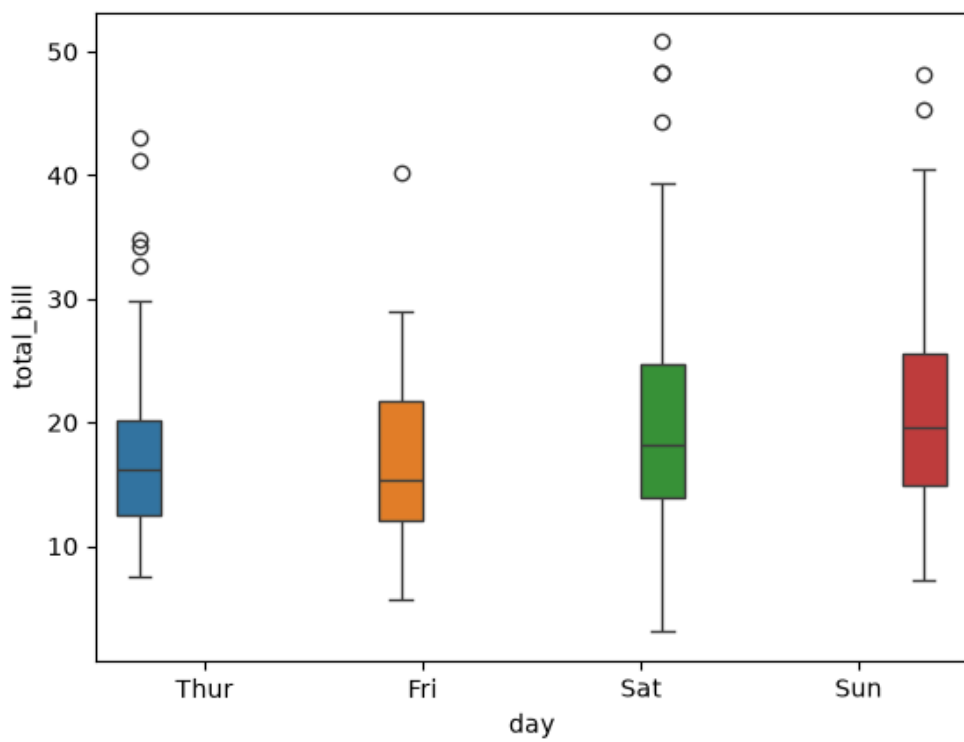
- Categorical Bar:



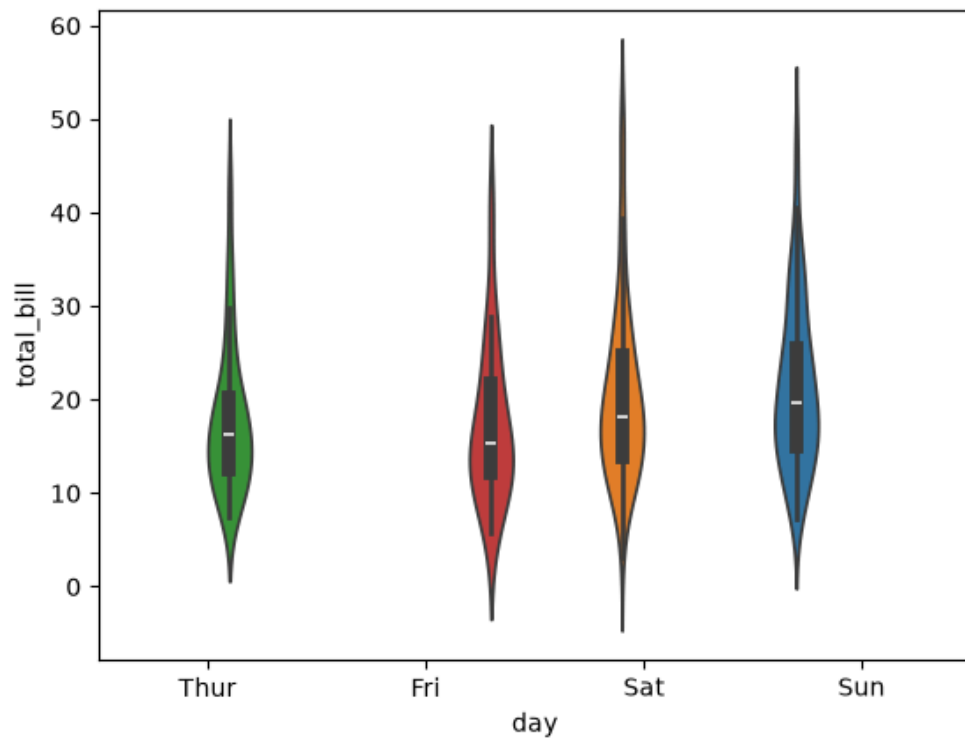
- Categorical Count:



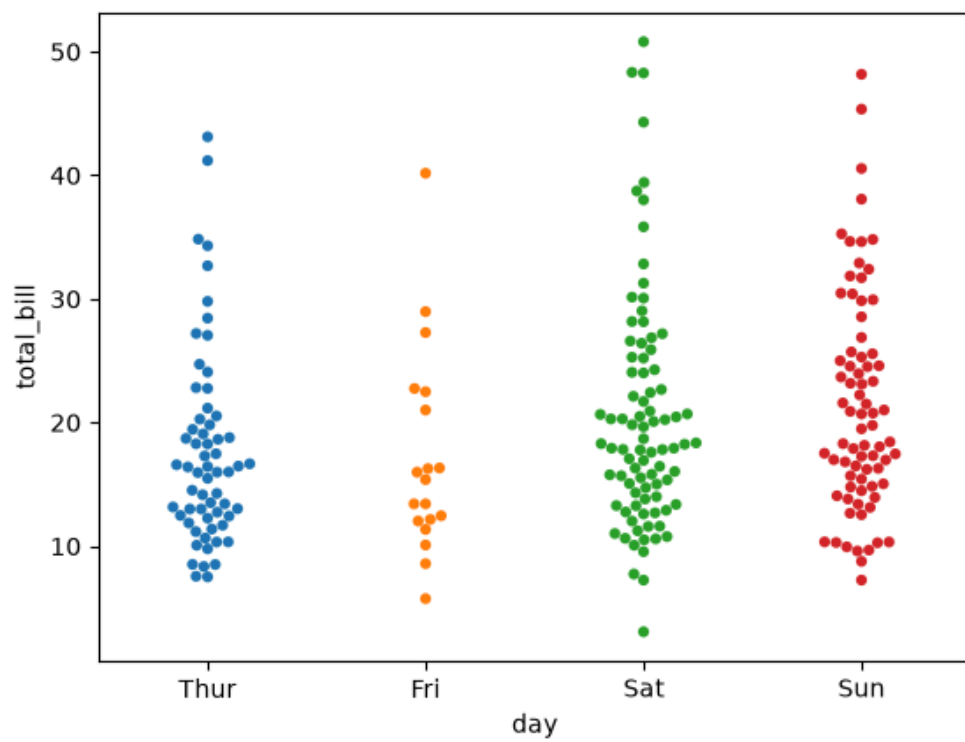
- Categorical Box:



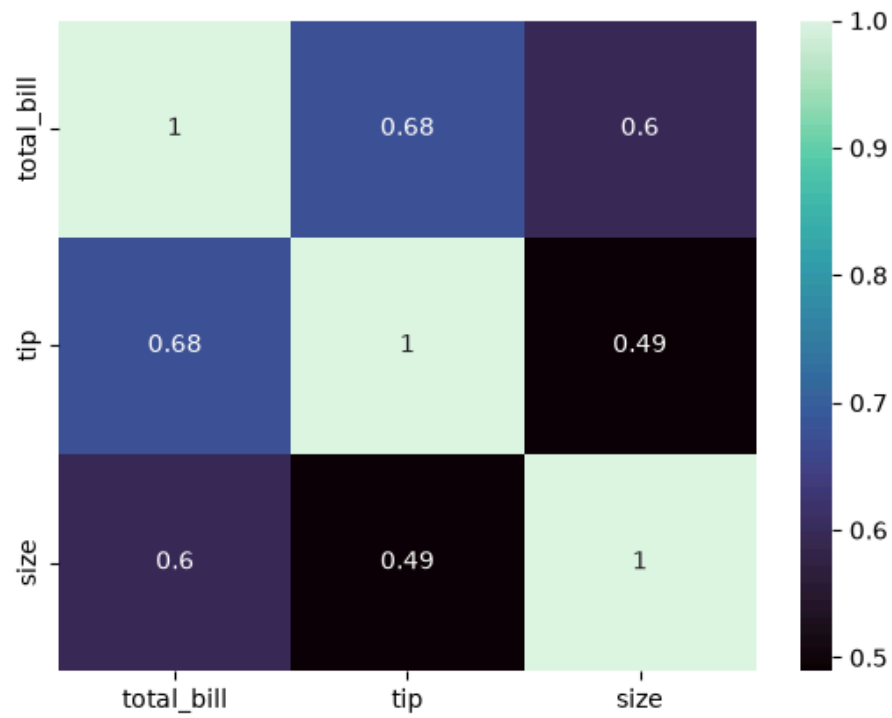
- Categorical Violin:



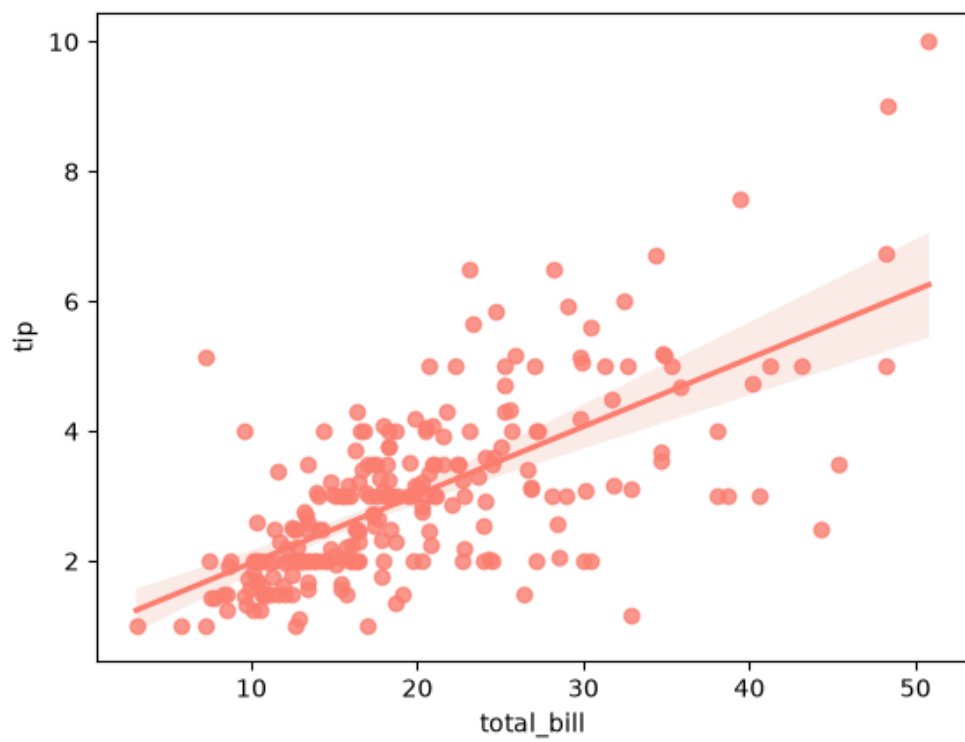
- Categorical Swarm:



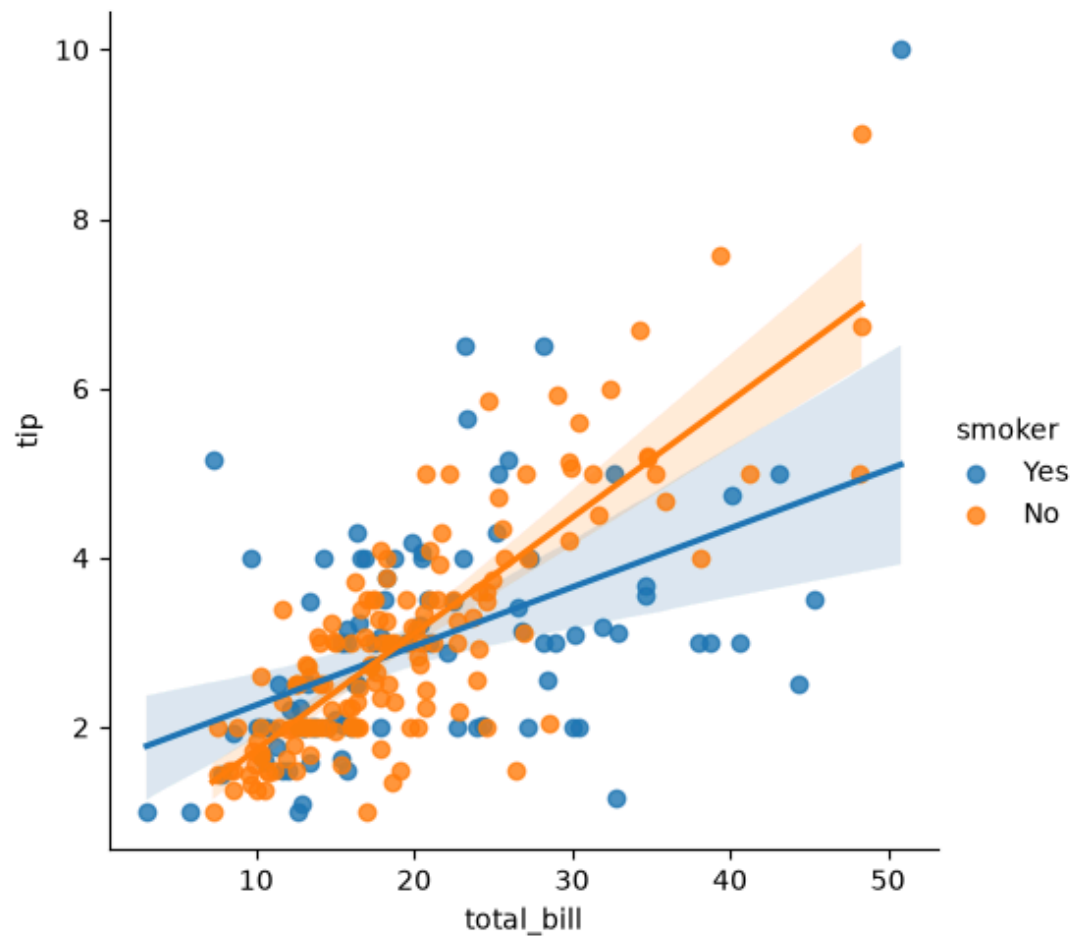
- Matrix Heatmap:



- Regression Regplot:

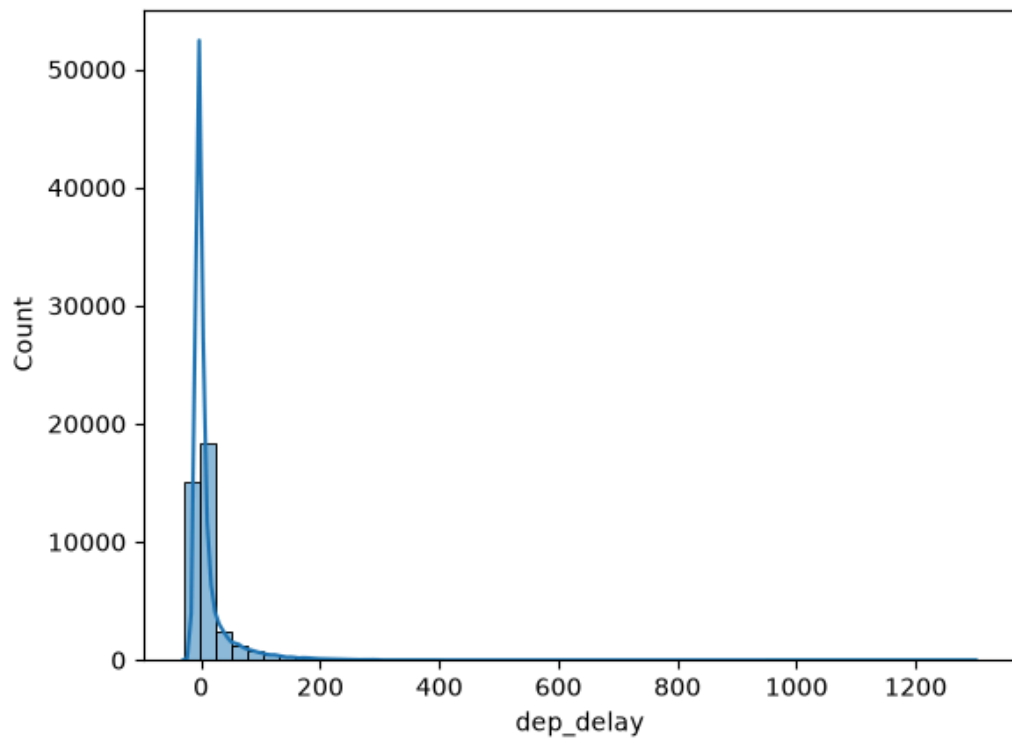


- Regression Lmplot:

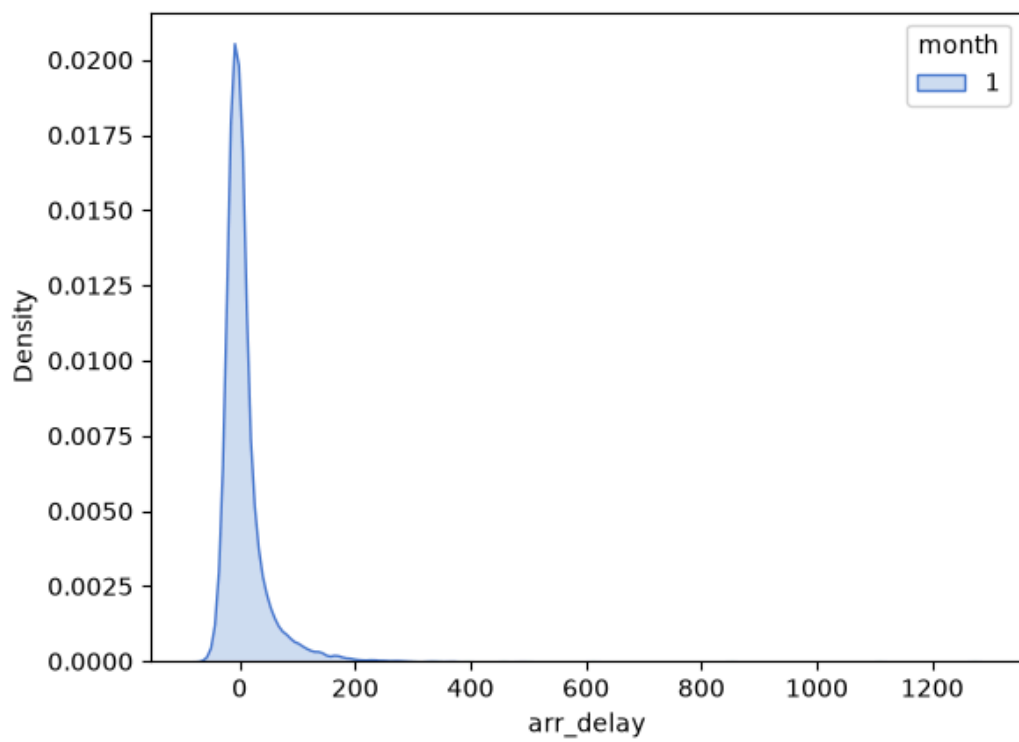


Visuals (Flights):

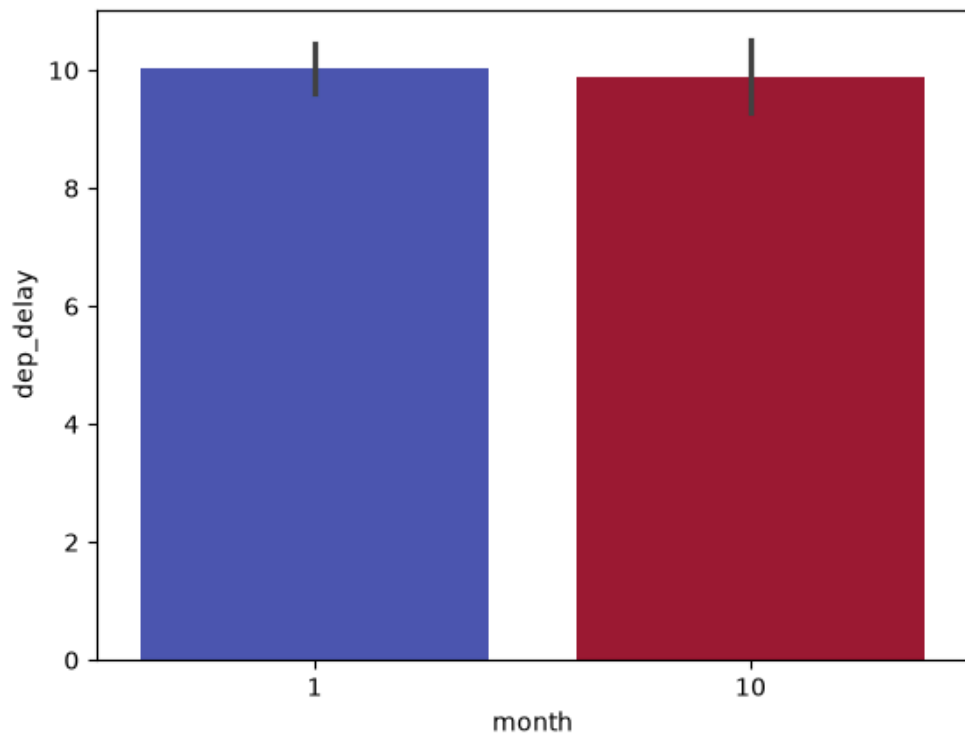
- Distribution Histplot:



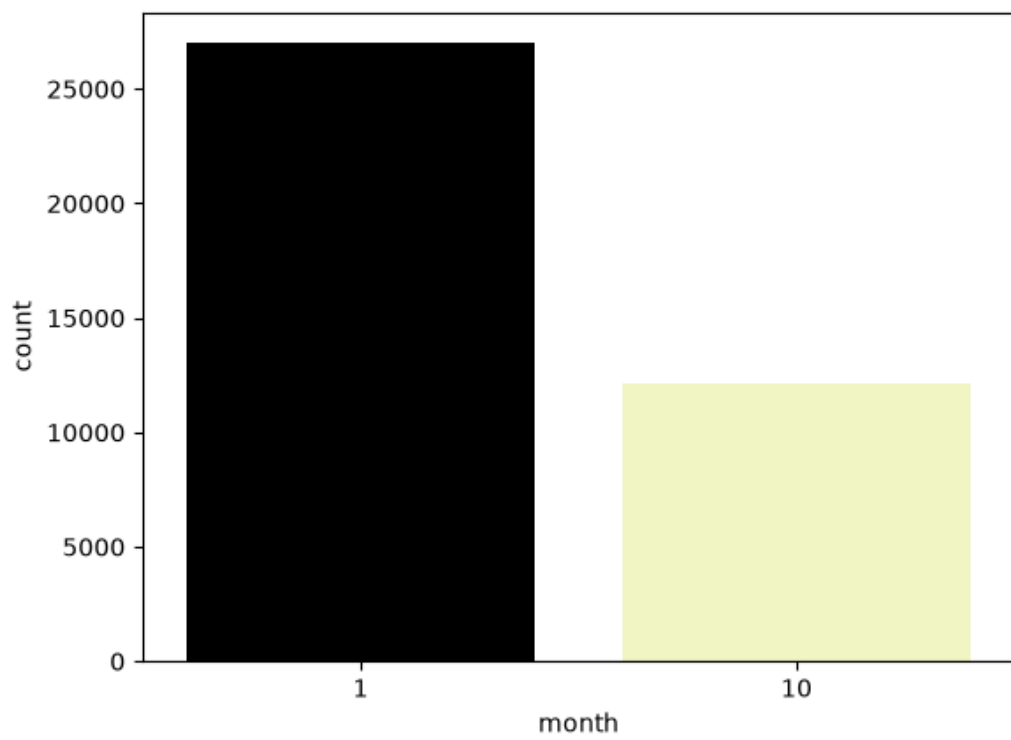
- Distribution KDE:



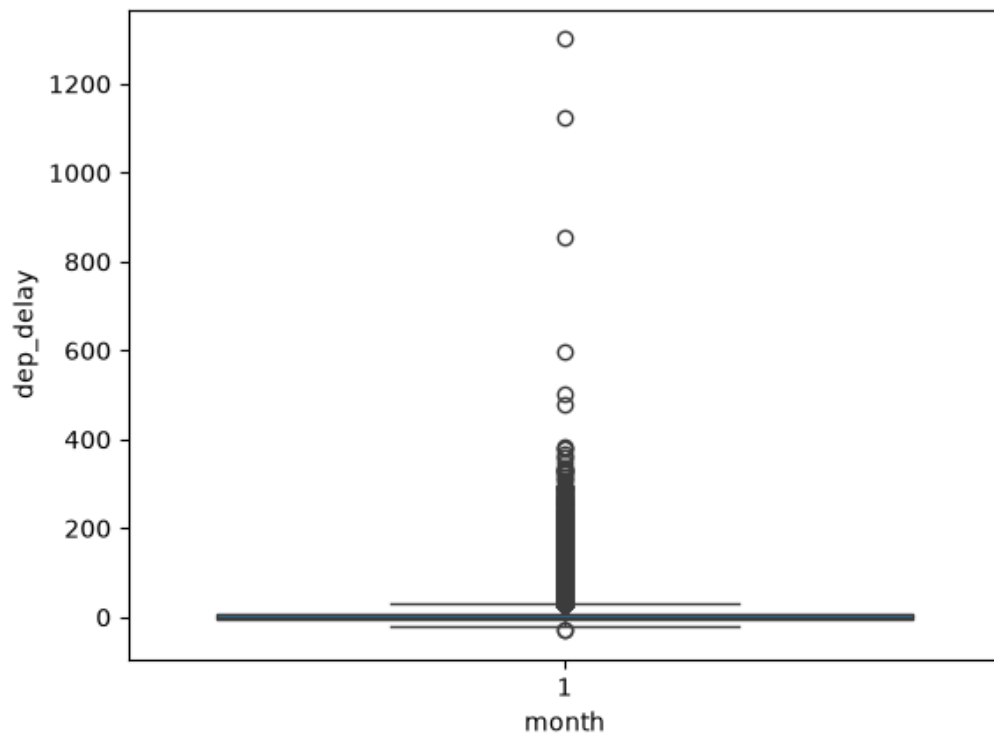
- Categorical Bar:



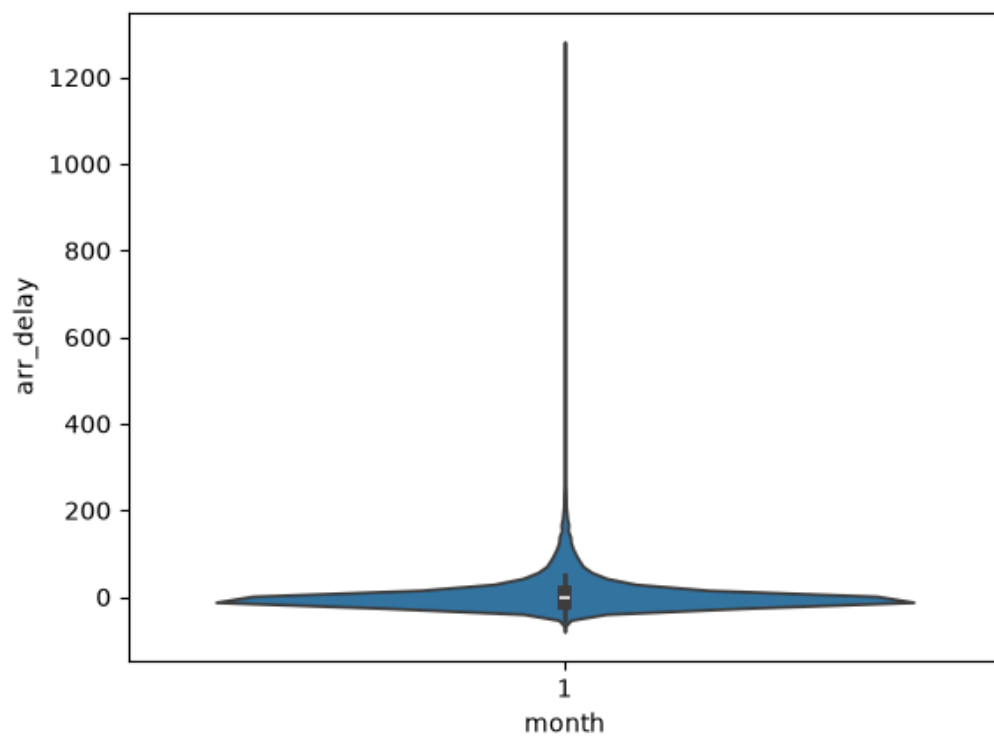
- Categorical Count:



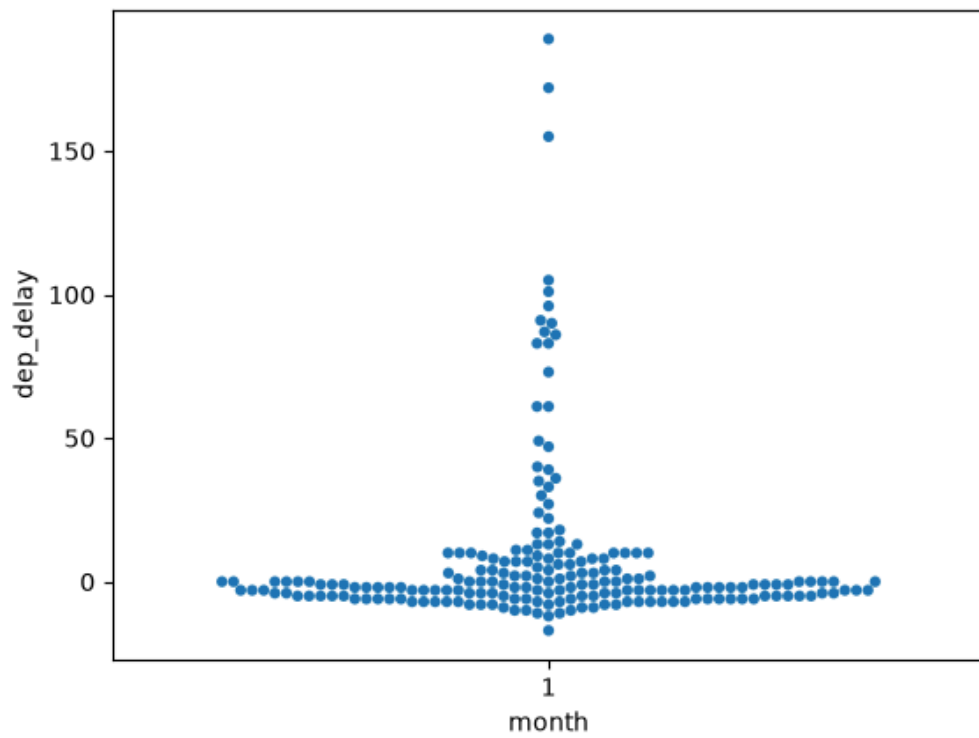
- Categorical Box:



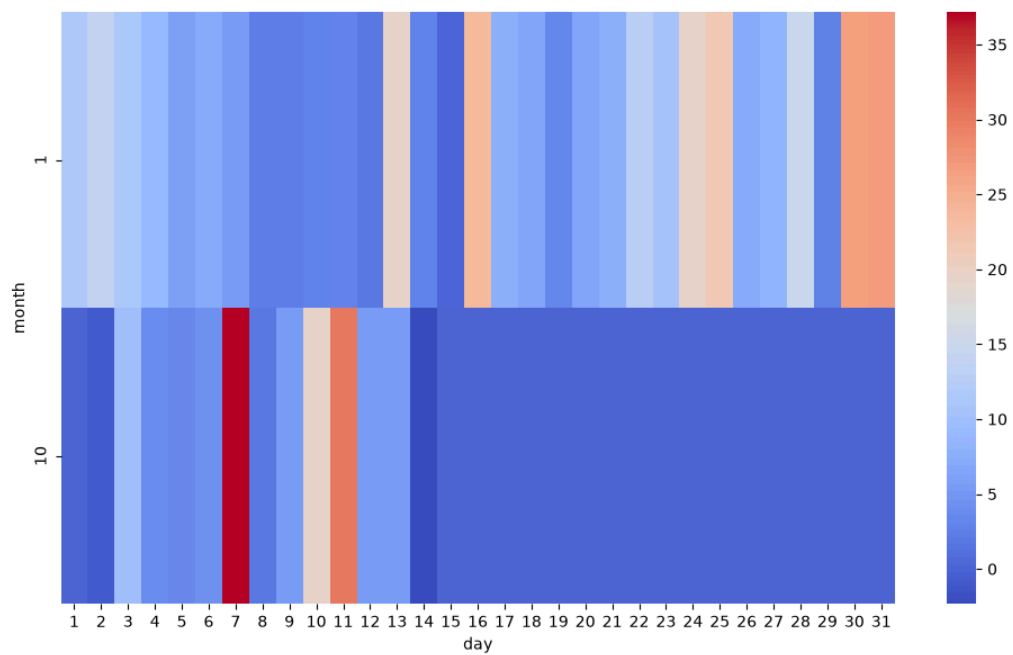
- Categorical Violin:



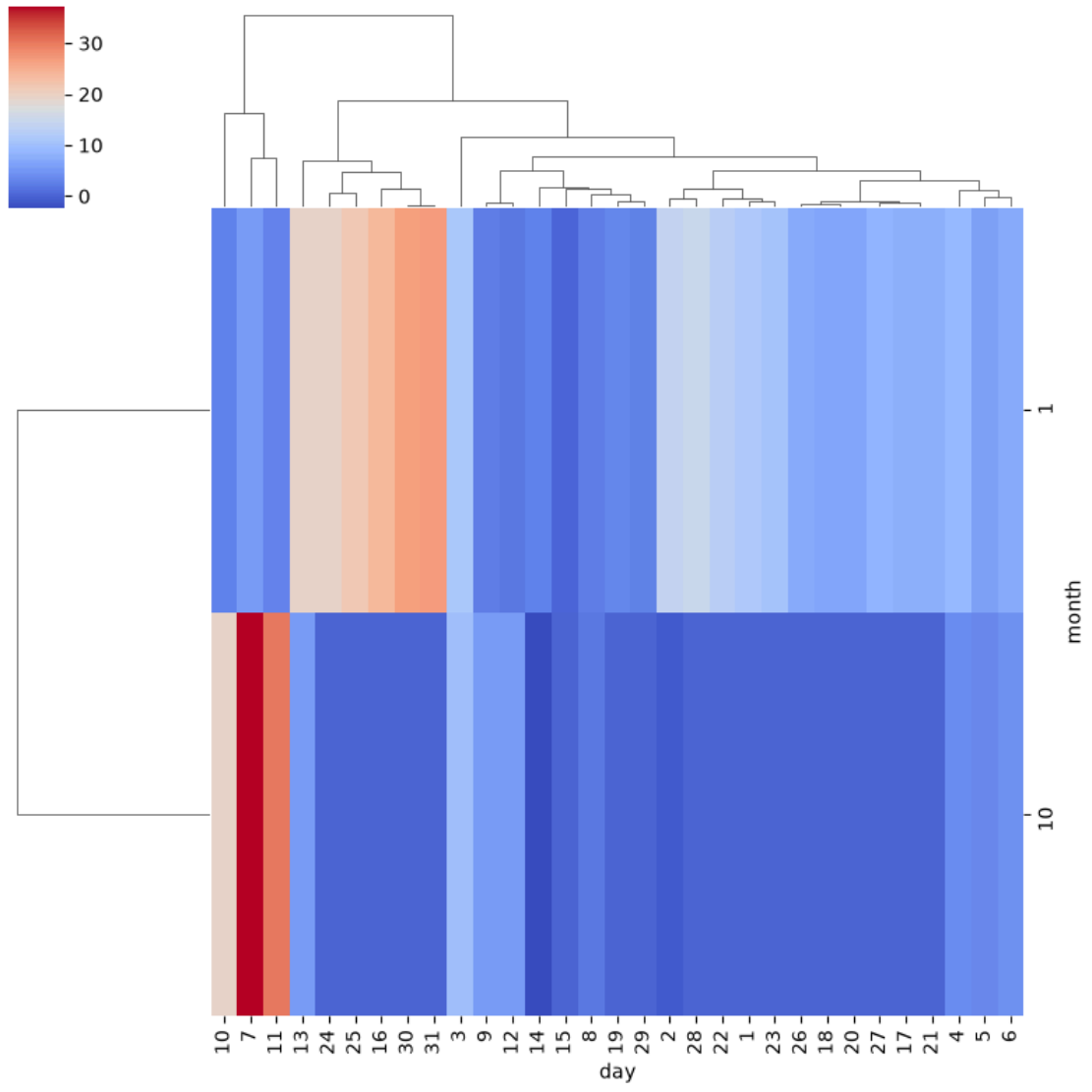
- Categorical Swarm:



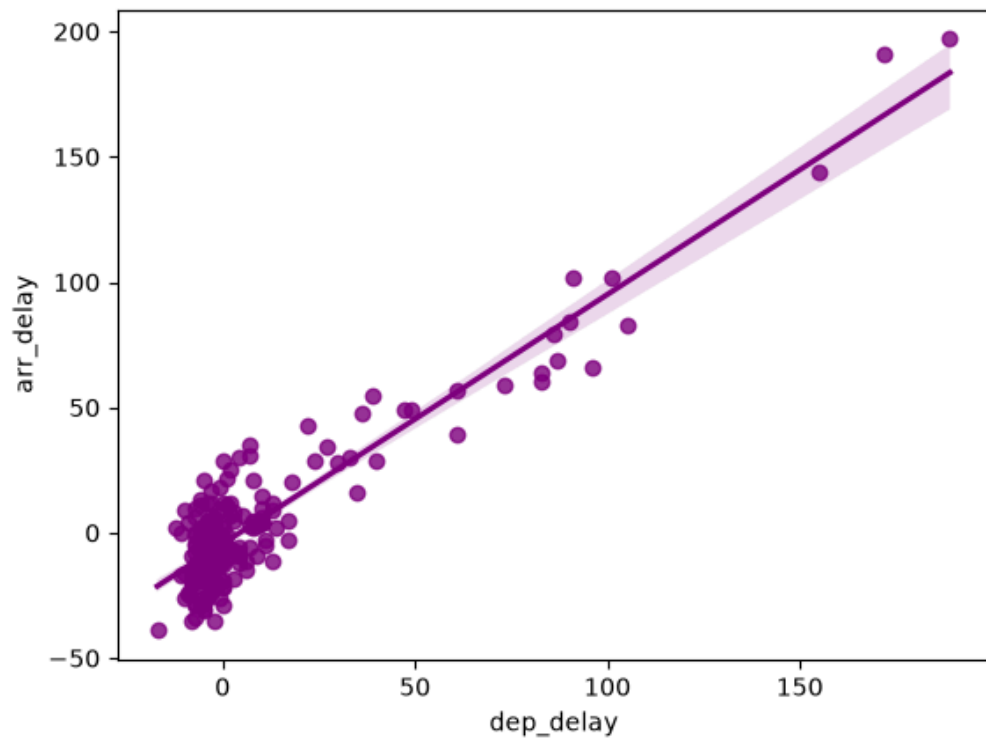
- Matrix Heatmap (Month vs Day Delay):



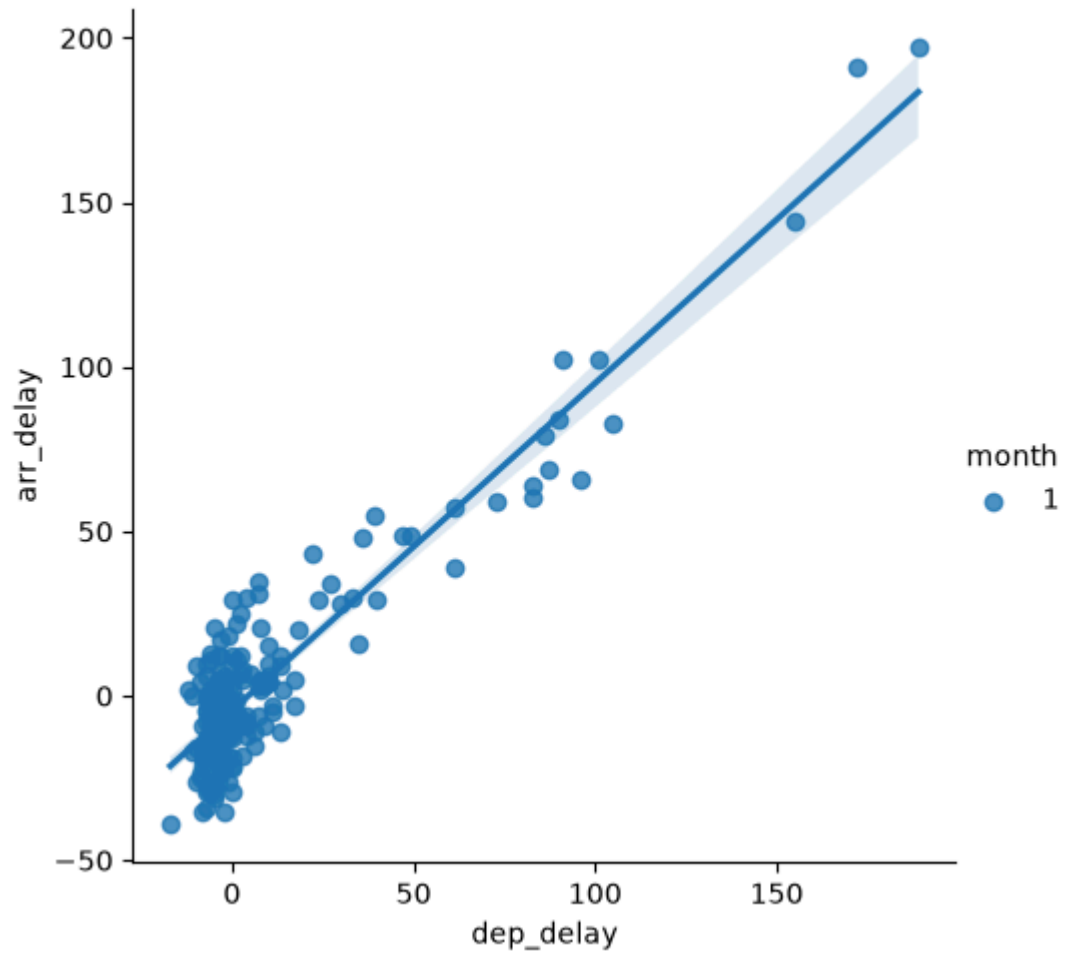
- **Matrix Clustermap (Month vs Day Delay):**



- Regression Regplot:

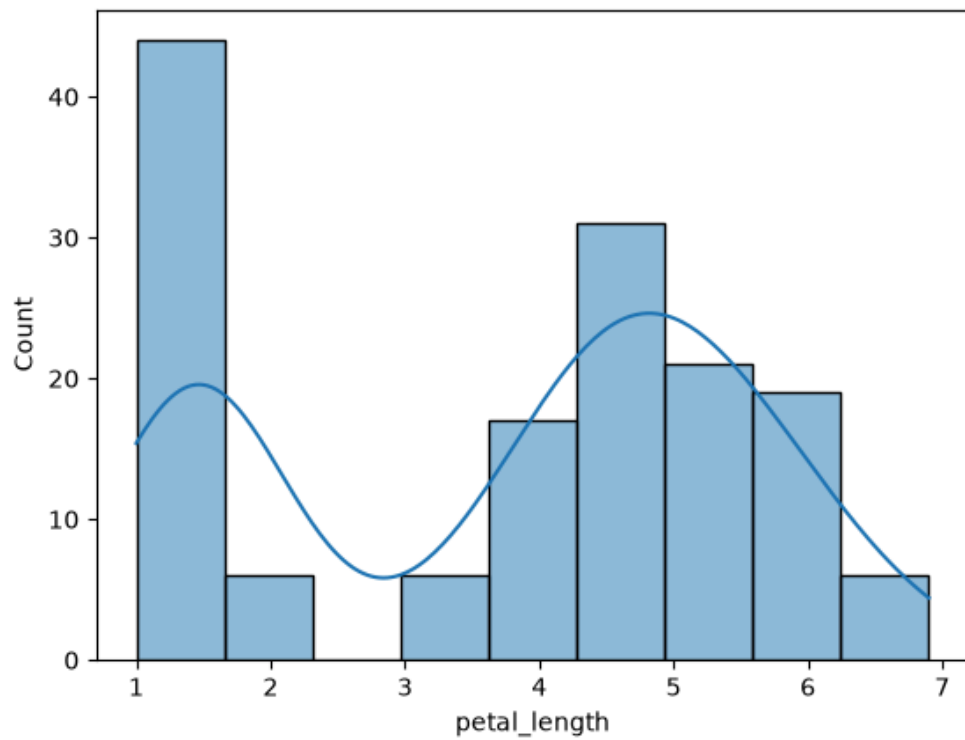


- Regression Lmplot:

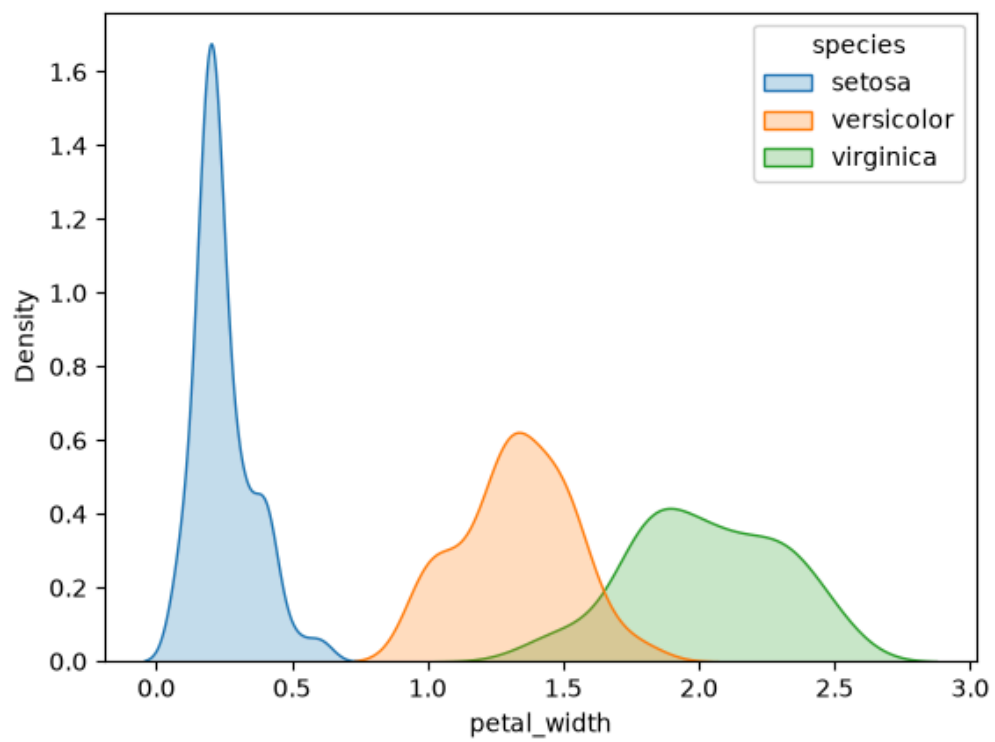


Visuals (Iris):

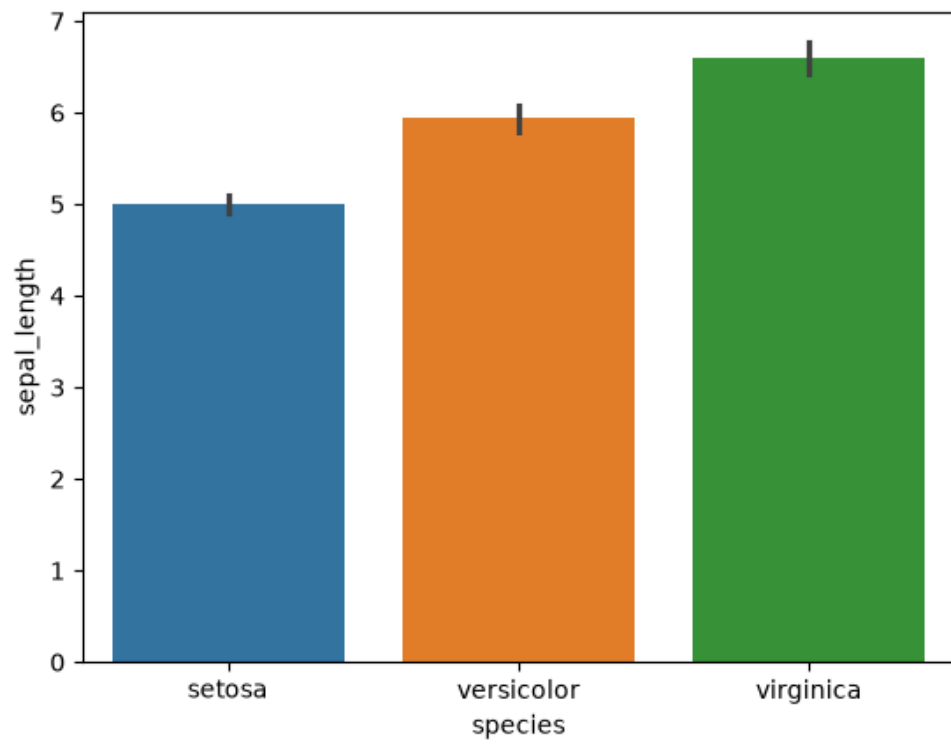
- Distribution Histplot:



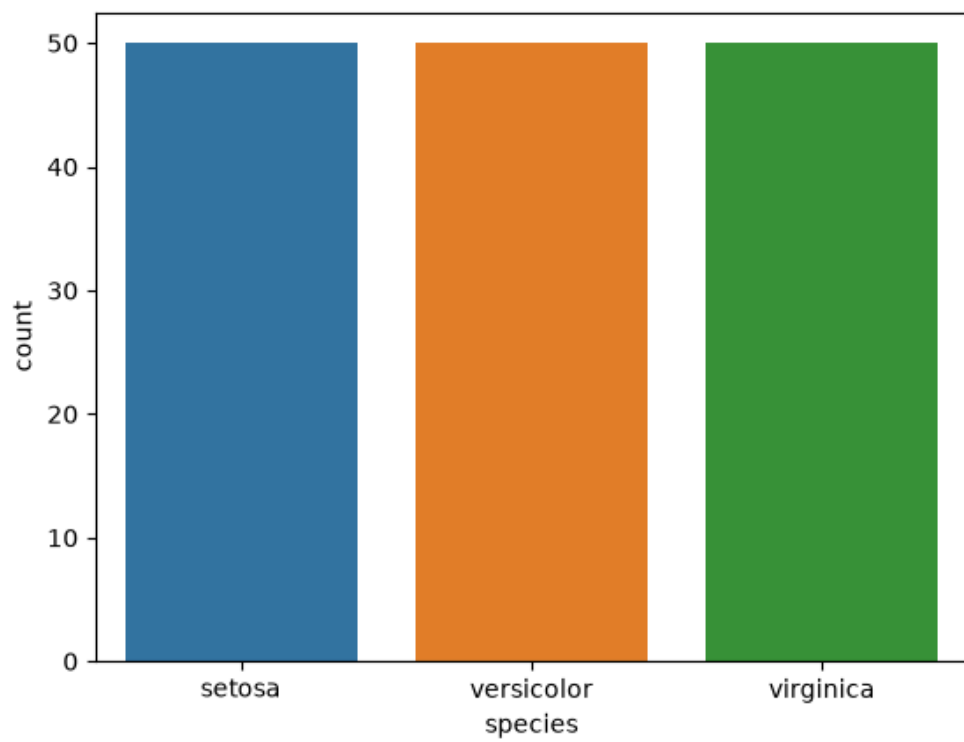
- Distribution KDE:



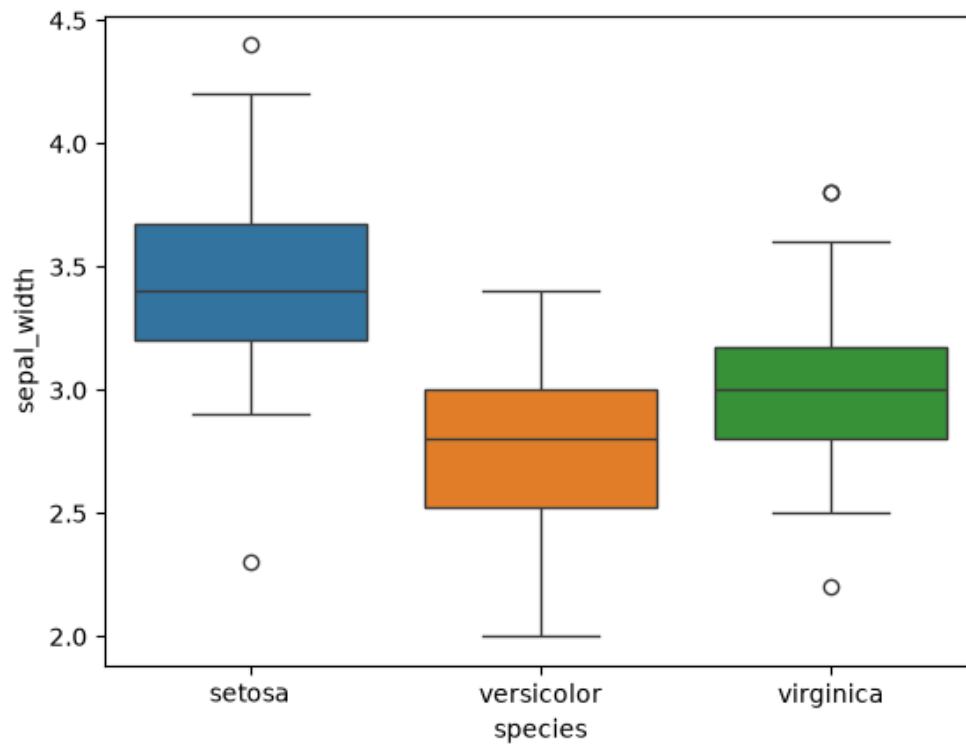
- Categorical Bar:



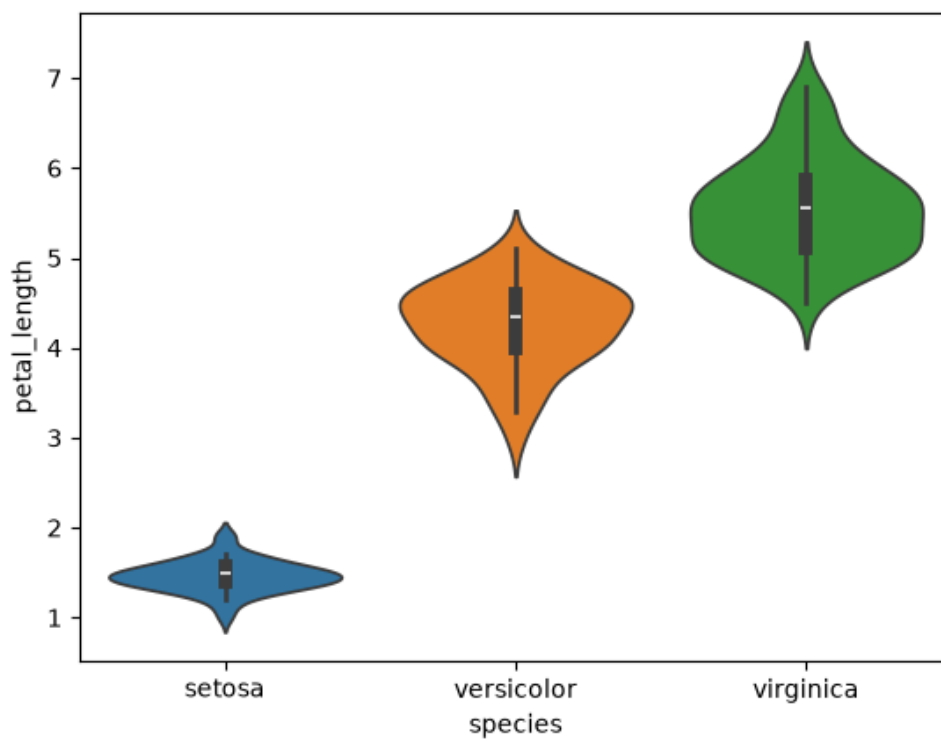
- Categorical Count:



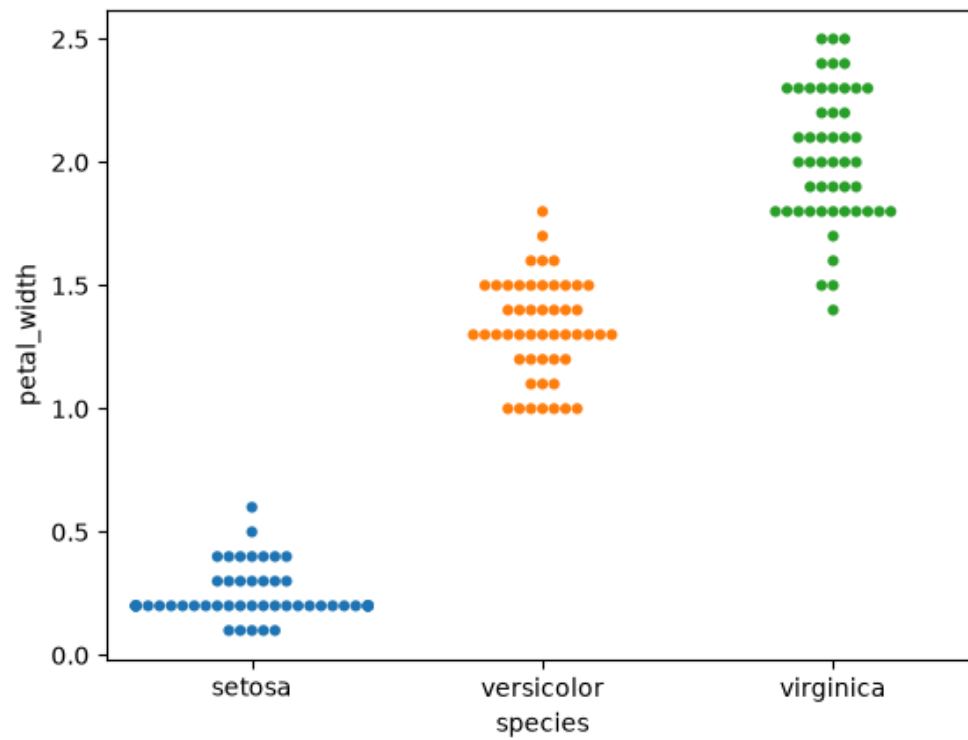
- Categorical Box:



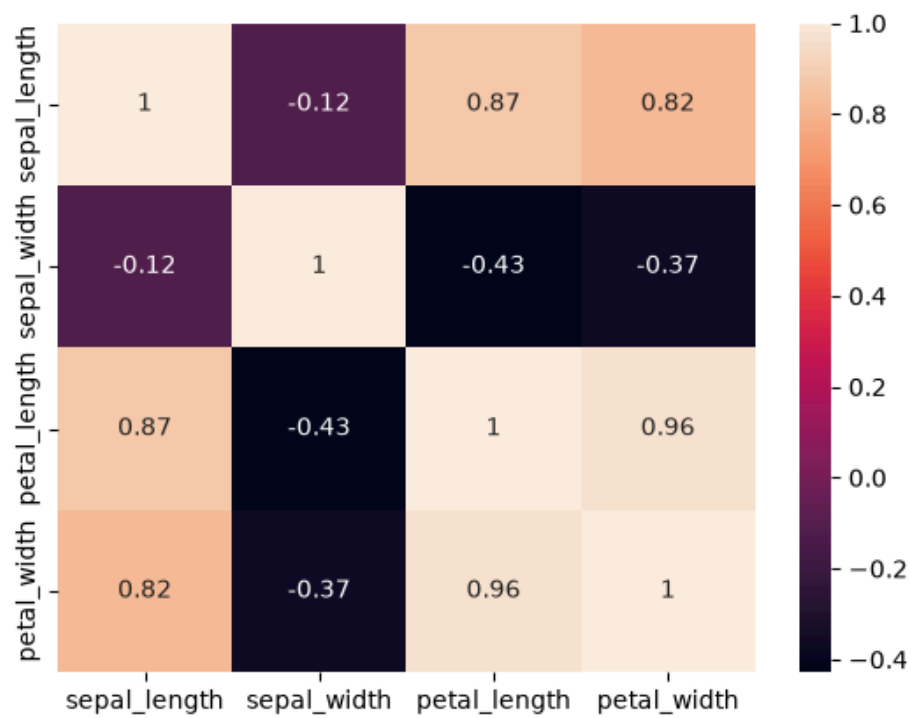
- Categorical Violin:



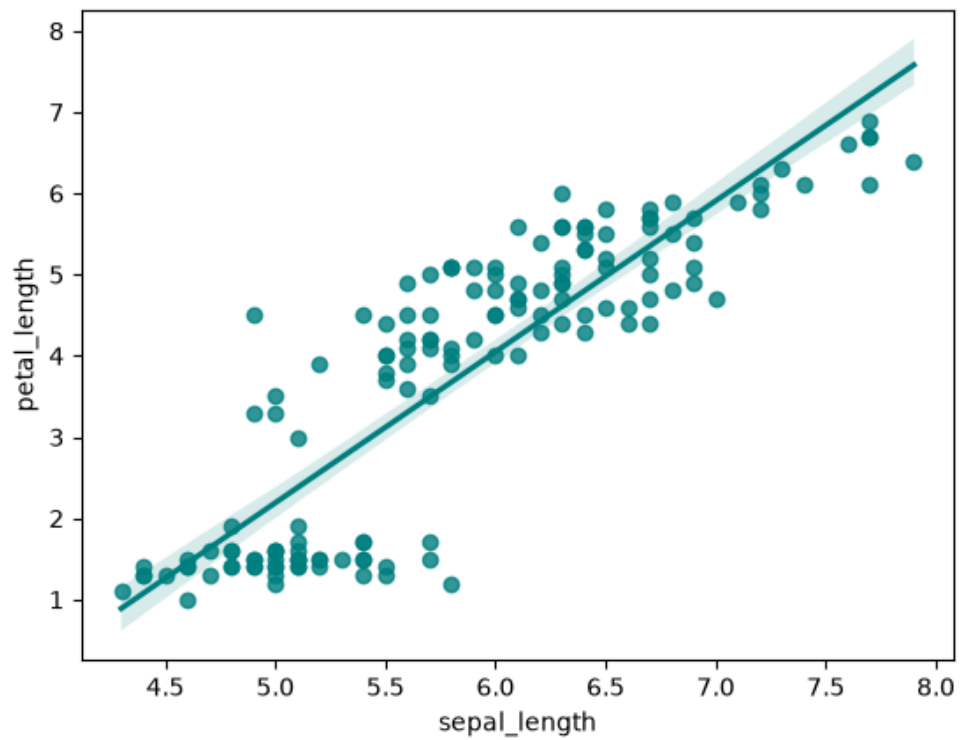
- Categorical Swarm:



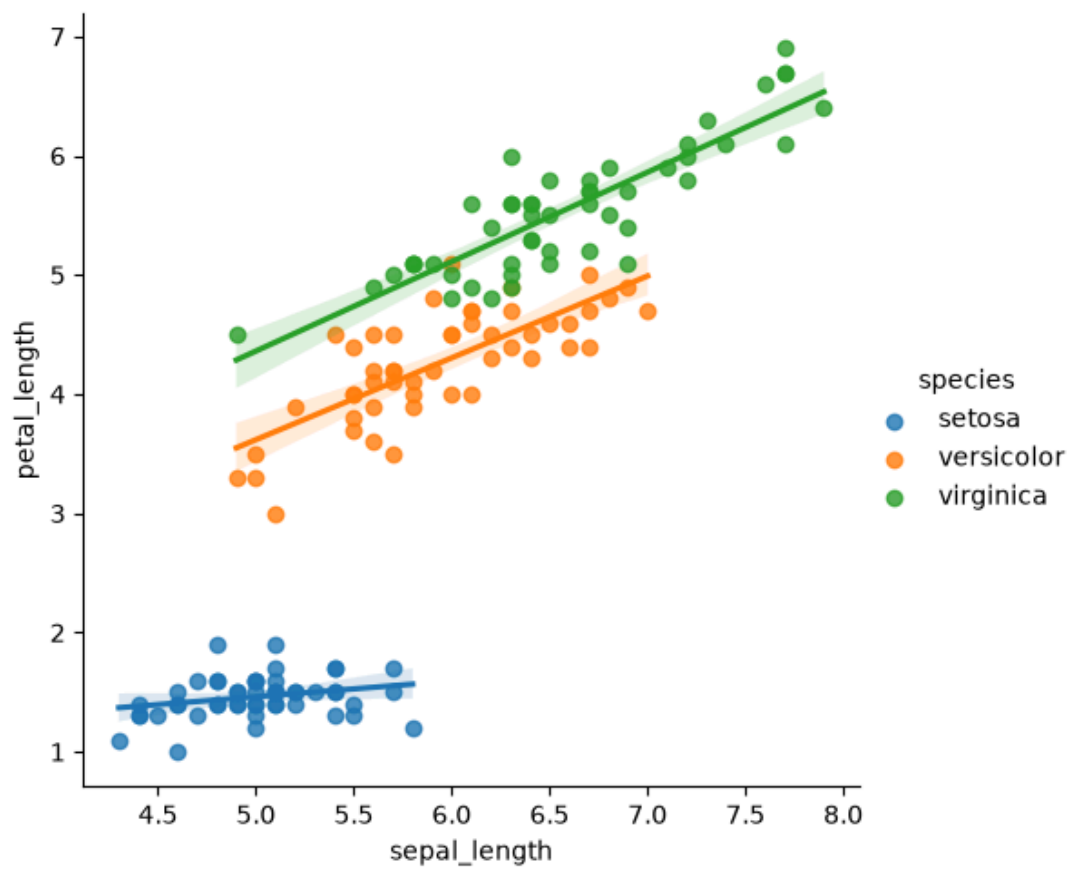
- **Matrix Heatmap:**



- Regression Regplot:



- Regression Lmplot:



Section 3: Grid, Style, and Color Customization

This section explores subplots using `PairGrid`, `FacetGrid`, and `JointGrid`, as well as Seaborn's custom styles and color palettes in `section3.py`.

Code Implementation (`section3.py`)

Code:

```
python

import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import os

os.makedirs('images', exist_ok=True)

# --- Grid Plots for Tips ---
tips_df = sns.load_dataset('tips')

g = sns.PairGrid(tips_df, hue='smoker')
g.map_diag(sns.histplot)
g.map_offdiag(sns.scatterplot)
g.savefig('images/tips_section3_pairgrid.png')
plt.close()

g = sns.FacetGrid(tips_df, col='time')
g.map(sns.scatterplot, 'total_bill', 'tip')
g.savefig('images/tips_section3_facetgrid.png')
plt.close()

g = sns.JointGrid(data=tips_df, x='total_bill', y='tip', hue='smoker')
g.plot(sns.scatterplot, sns.histplot)
g.savefig('images/tips_section3_jointgrid.png')
plt.close()

sns.set_style('darkgrid')
plt.figure()
sns.boxplot(data=tips_df, x='day', y='total_bill', hue='day', legend=False,
palette='Set2')
plt.title('Darkgrid Boxplot')
plt.savefig('images/tips_section3_style_darkgrid.png')
plt.close()
```

```

sns.set_style('whitegrid')
plt.figure()
sns.boxplot(data=tips_df, x='day', y='total_bill', hue='day', legend=False,
palette='pastel')
plt.title('Whitegrid Boxplot')
plt.savefig('images/tips_section3_style_whitegrid.png')
plt.close()

sns.set_style('ticks')
plt.figure()
sns.boxplot(data=tips_df, x='day', y='total_bill', hue='day', legend=False,
palette='muted')
plt.title('Ticks Boxplot')
plt.savefig('images/tips_section3_style_ticks.png')
plt.close()

# --- Grid Plots for Flights ---
flights_df = pd.read_csv('../flights.csv')
flights_df['dep_delay'] =
flights_df['dep_delay'].fillna(flights_df['dep_delay'].mean())
flights_df['arr_delay'] =
flights_df['arr_delay'].fillna(flights_df['arr_delay'].mean())

flights_subset = flights_df[flights_df['month'].isin([1, 2, 3])]
flights_sample = flights_subset.sample(200, random_state=42)

g = sns.PairGrid(flights_sample[['dep_delay', 'arr_delay', 'month']], hue='month',
palette='Dark2')
g.map_diag(sns.histplot)
g.map_offdiag(sns.scatterplot)
g.savefig('images/flights_section3_pairgrid.png')
plt.close()

g = sns.FacetGrid(flights_subset, col='month', height=4, aspect=1)
g.map(sns.scatterplot, 'dep_delay', 'arr_delay', alpha=0.5, color='orange')
g.savefig('images/flights_section3_facetgrid.png')
plt.close()

g = sns.JointGrid(data=flights_sample, x='dep_delay', y='arr_delay', hue='month',
palette='Dark2')
g.plot(sns.scatterplot, sns.histplot)

```

```

g.savefig('images/flights_section3_jointgrid.png')
plt.close()

sns.set_style('darkgrid')
plt.figure()
sns.boxplot(data=flights_subset, x='month', y='dep_delay', hue='month',
            legend=False, palette='Set2')
plt.savefig('images/flights_section3_style_darkgrid.png')
plt.close()

sns.set_style('whitegrid')
plt.figure()
sns.boxplot(data=flights_subset, x='month', y='dep_delay', hue='month',
            legend=False, palette='pastel')
plt.savefig('images/flights_section3_style_whitegrid.png')
plt.close()

sns.set_style('ticks')
plt.figure()
sns.boxplot(data=flights_subset, x='month', y='dep_delay', hue='month',
            legend=False, palette='muted')
plt.savefig('images/flights_section3_style_ticks.png')
plt.close()

# --- Grid Plots for Iris ---
iris_df = sns.load_dataset('iris')

g = sns.PairGrid(iris_df, hue='species')
g.map_diag(sns.histplot)
g.map_offdiag(sns.scatterplot)
g.savefig('images/section3_pairgrid.png')
plt.close()

g = sns.FacetGrid(iris_df, col='species')
g.map(sns.scatterplot, 'sepal_length', 'sepal_width')
g.savefig('images/section3_facetgrid.png')
plt.close()

g = sns.JointGrid(data=iris_df, x='petal_length', y='petal_width', hue='species')
g.plot(sns.scatterplot, sns.histplot)
g.savefig('images/section3_jointgrid.png')
plt.close()

```

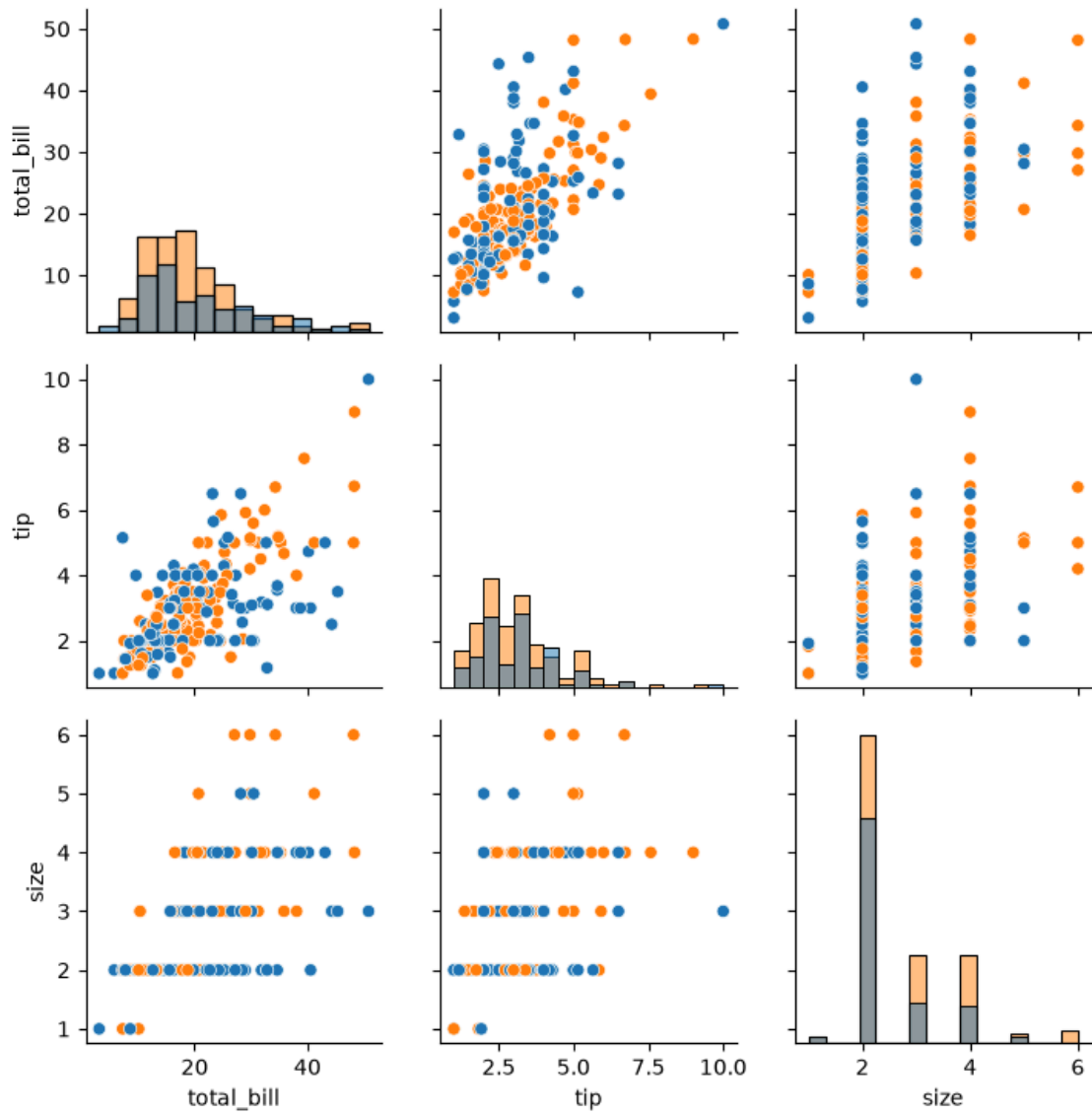
```
sns.set_style('darkgrid')
plt.figure()
sns.boxplot(data=iris_df, x='species', y='sepal_length', hue='species',
legend=False, palette='Set2')
plt.savefig('images/section3_style_darkgrid.png')
plt.close()

sns.set_style('whitegrid')
plt.figure()
sns.boxplot(data=iris_df, x='species', y='sepal_length', hue='species',
legend=False, palette='pastel')
plt.savefig('images/section3_style_whitegrid.png')
plt.close()

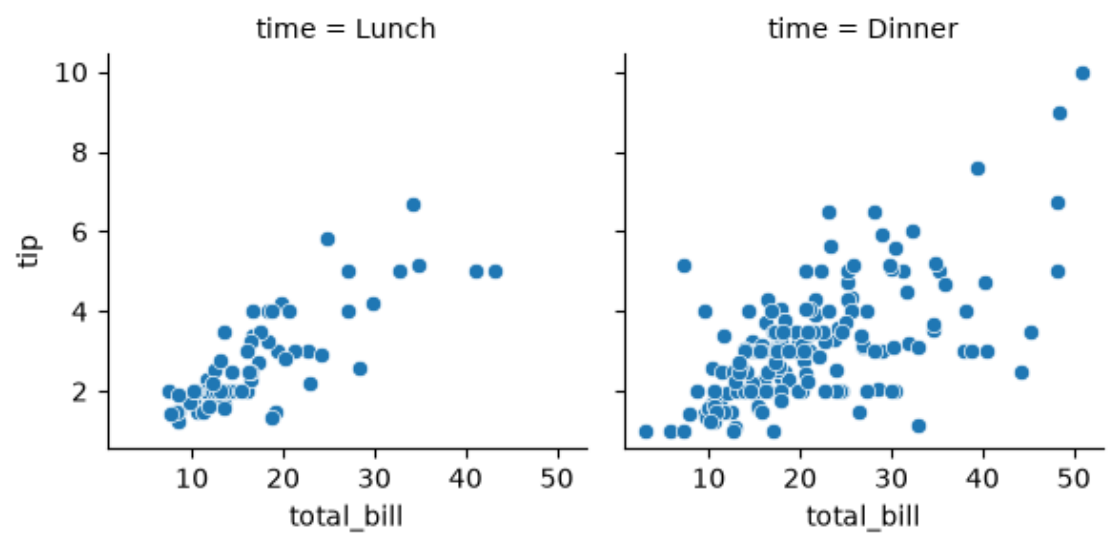
sns.set_style('ticks')
plt.figure()
sns.boxplot(data=iris_df, x='species', y='sepal_length', hue='species',
legend=False, palette='muted')
plt.savefig('images/section3_style_ticks.png')
plt.close()
```

Visuals (Tips):

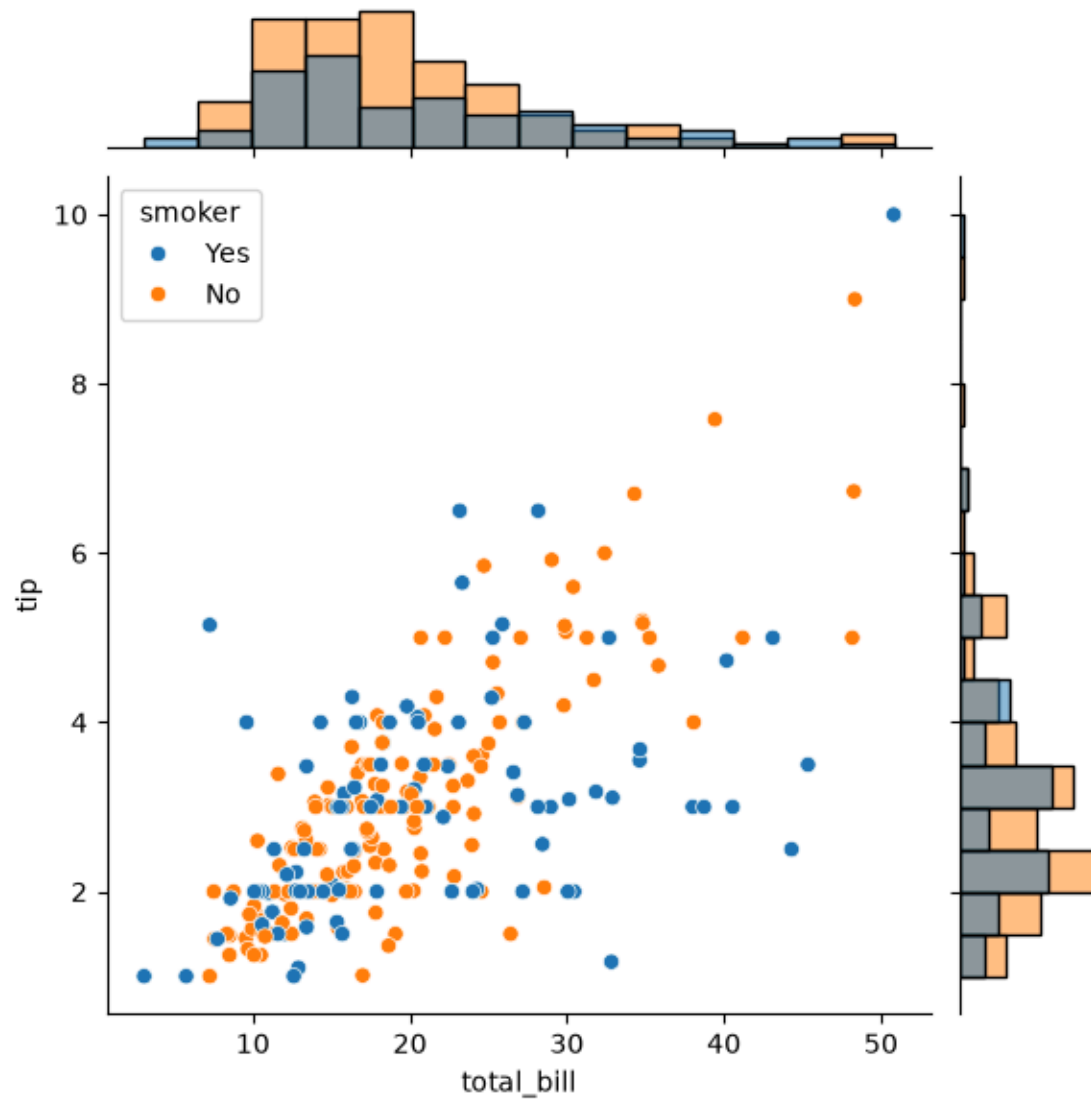
- PairGrid:



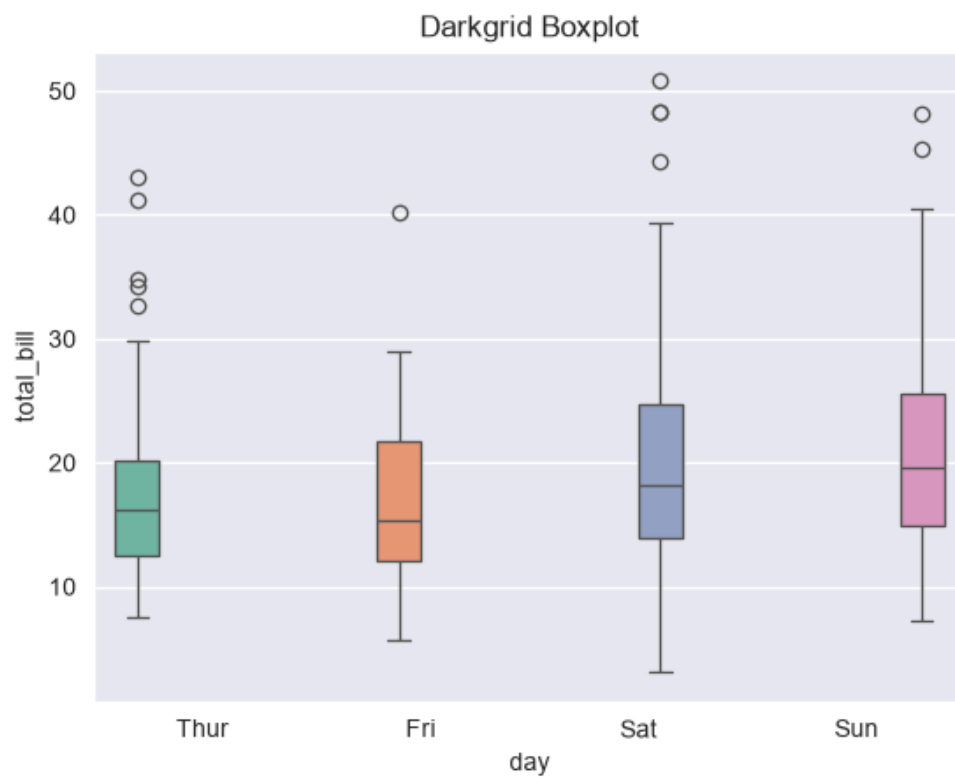
- FacetGrid:



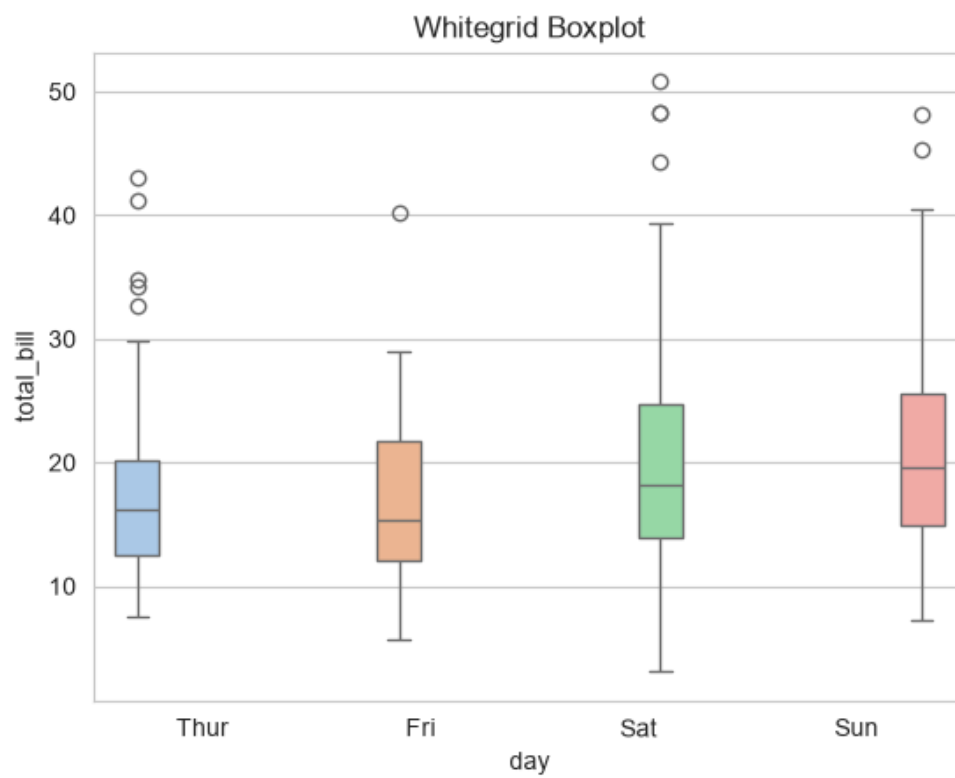
- JointGrid:



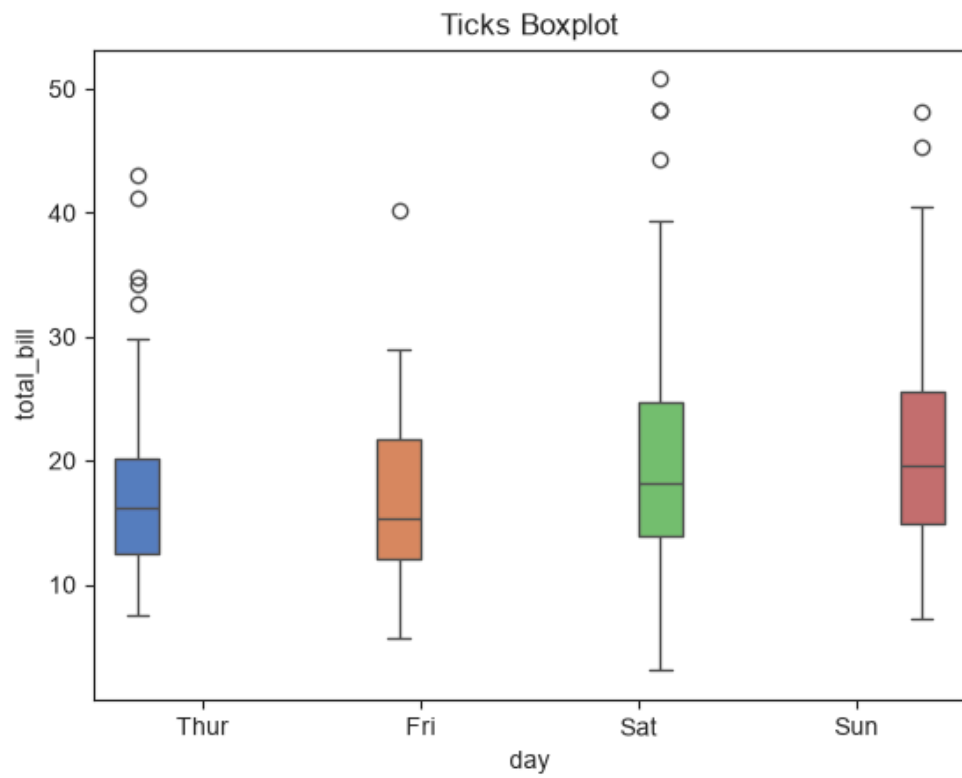
- Darkgrid Style + Viridis:



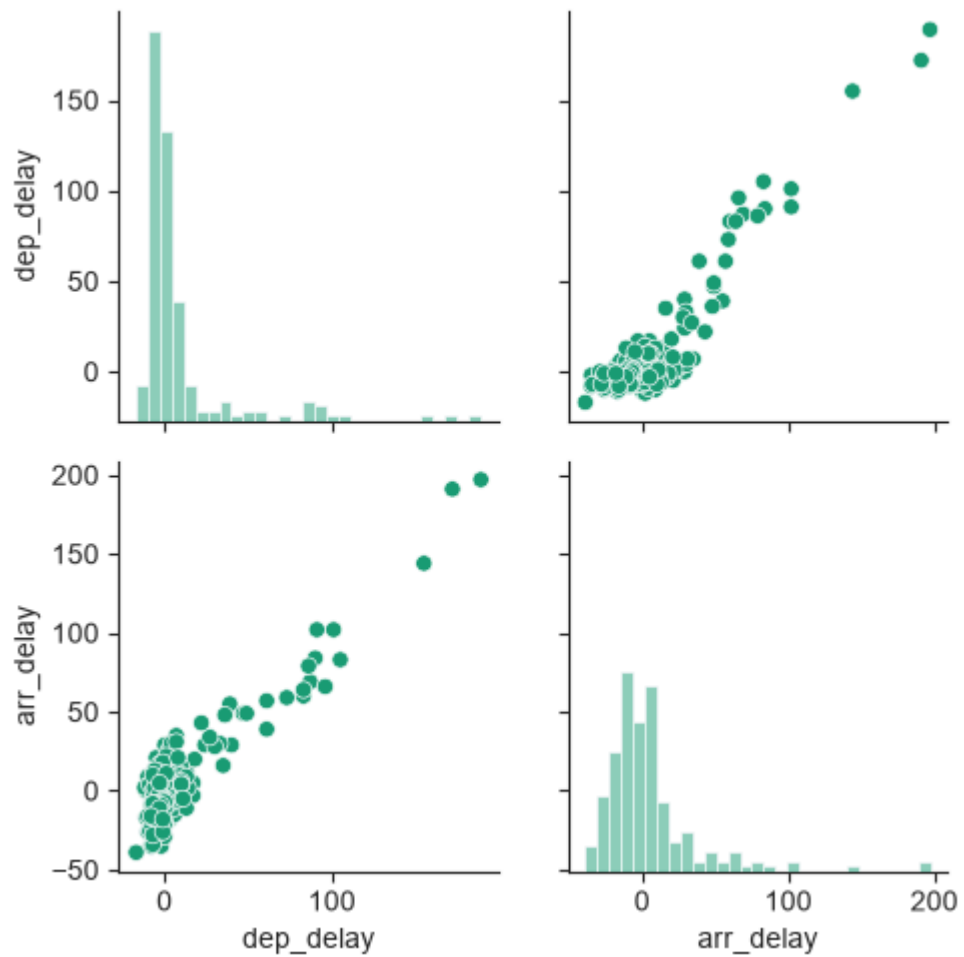
- Whitegrid Style + Plasma:



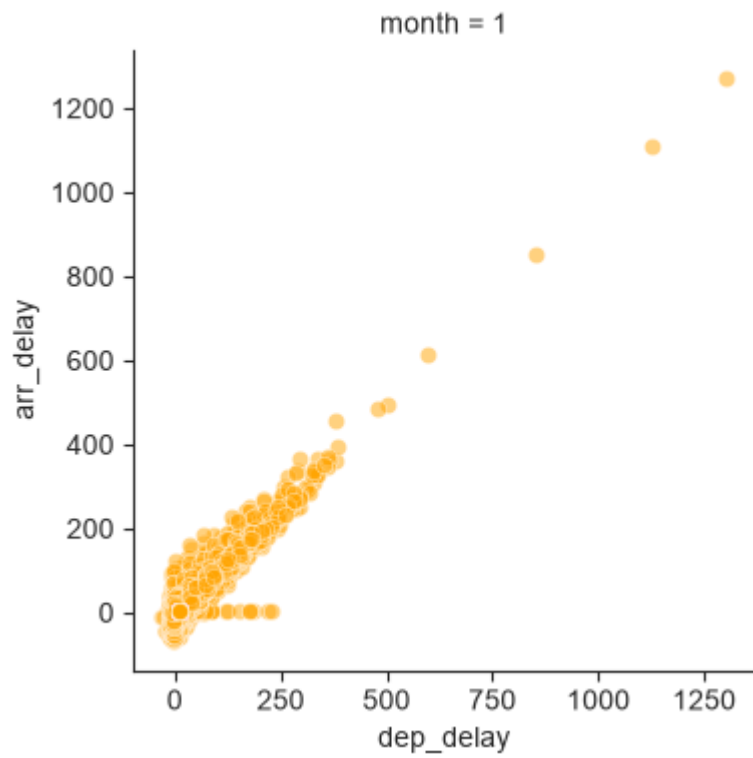
- Ticks Style + Muted:



Visuals (Flights):

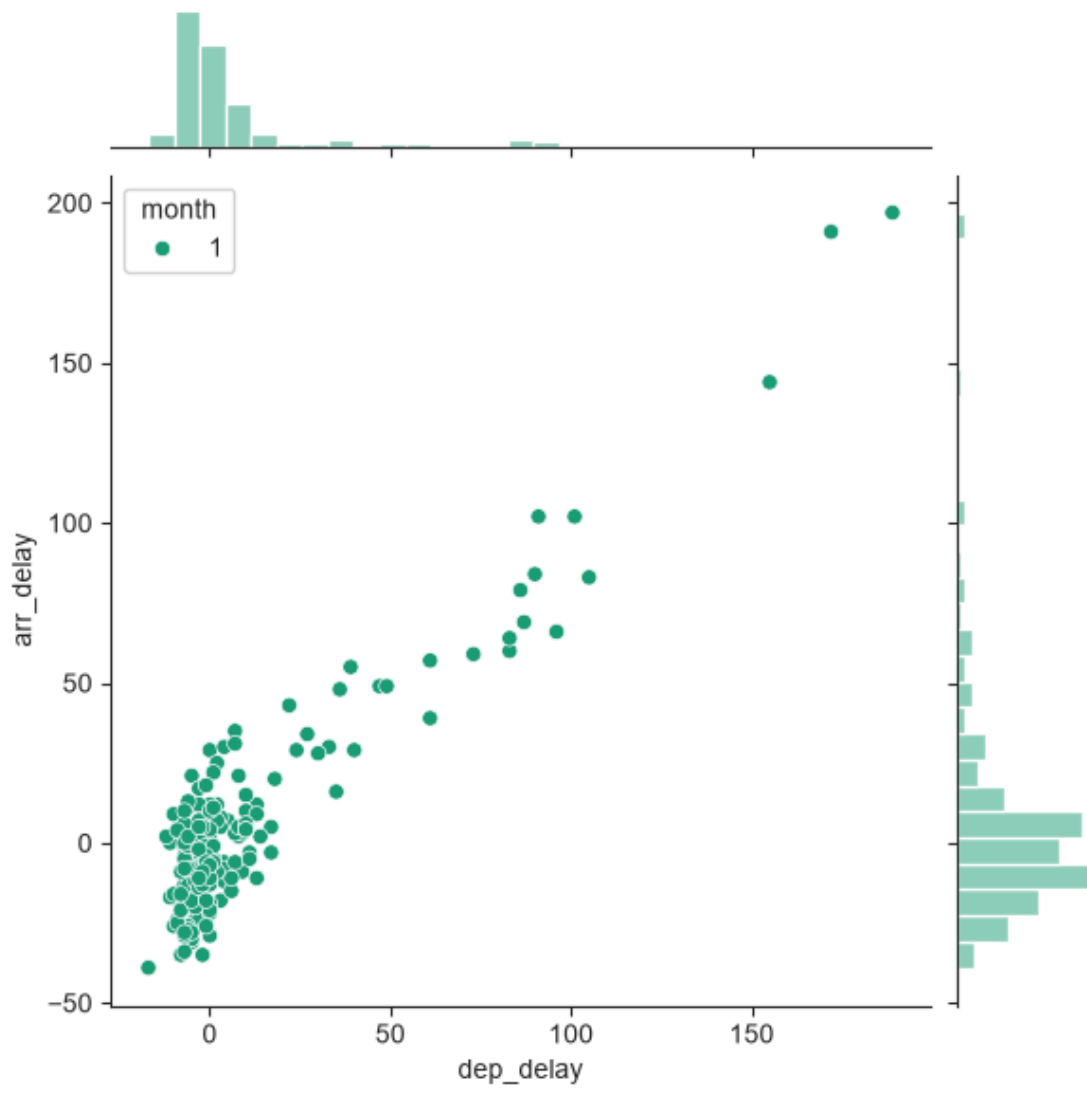


- PairGrid:

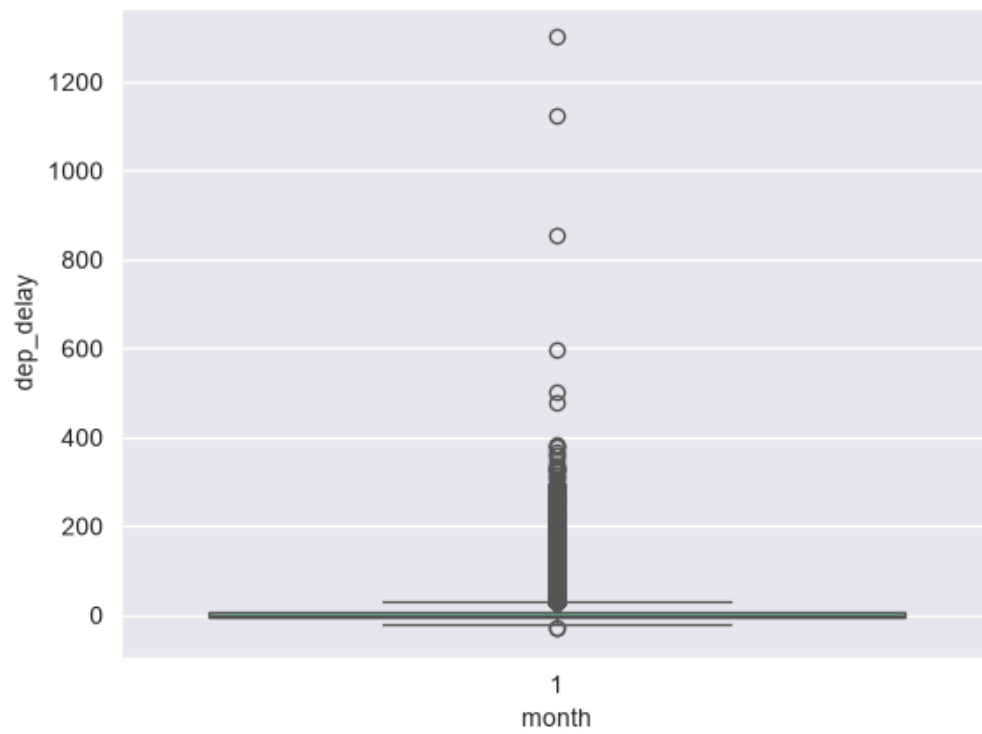


- FacetGrid:

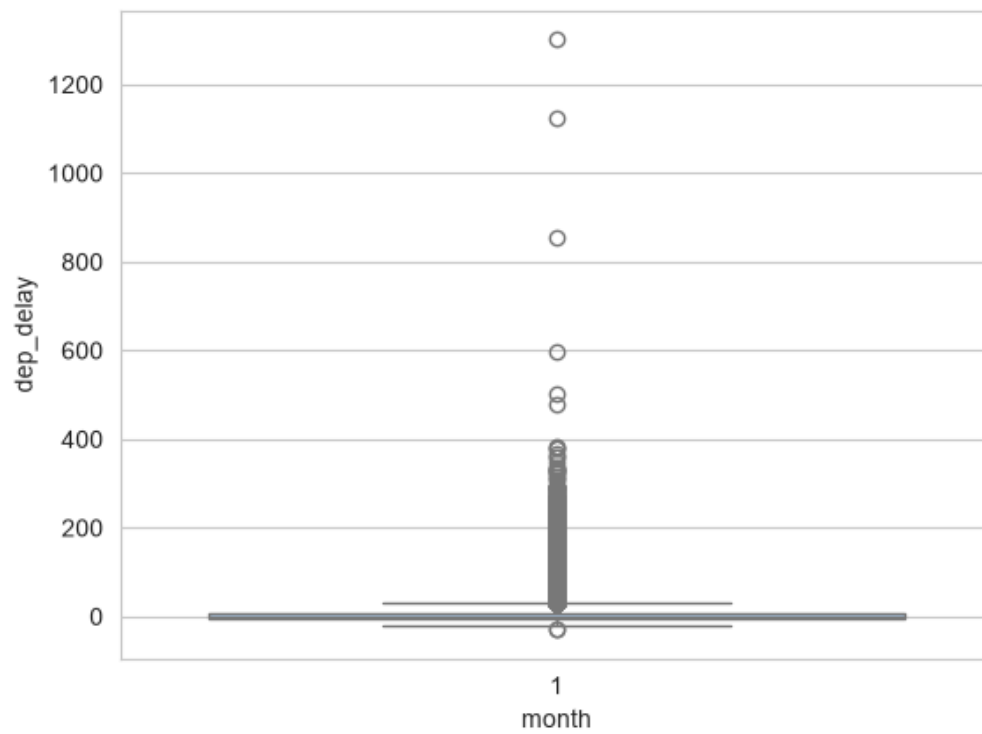
- JointGrid:



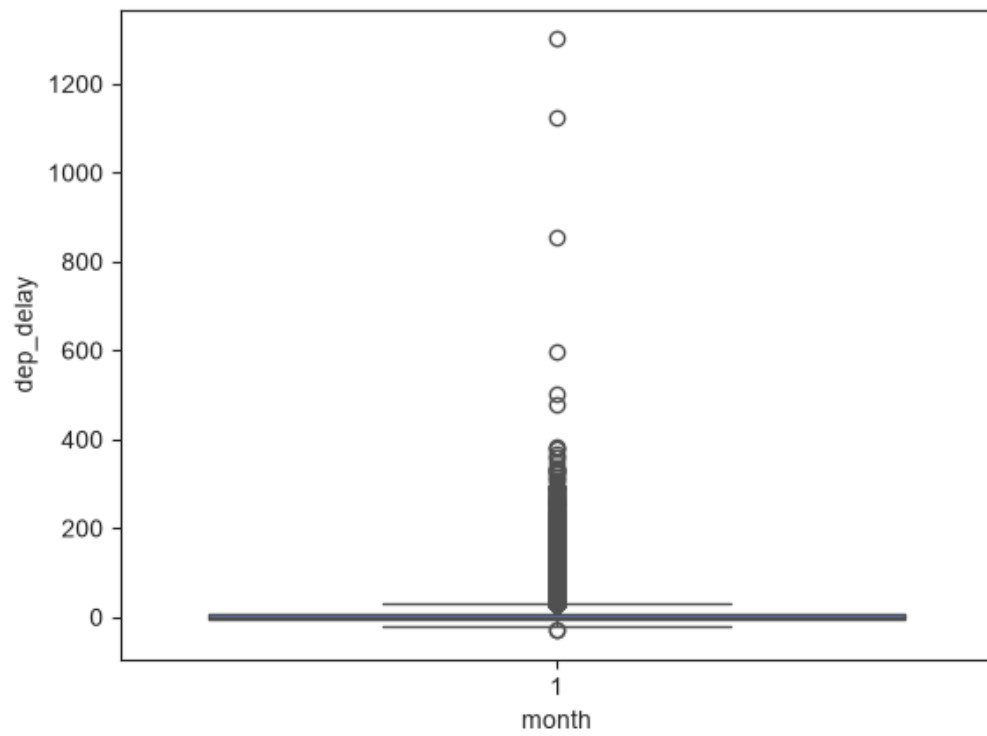
- Darkgrid Style + Viridis:



- Whitegrid Style + Plasma:

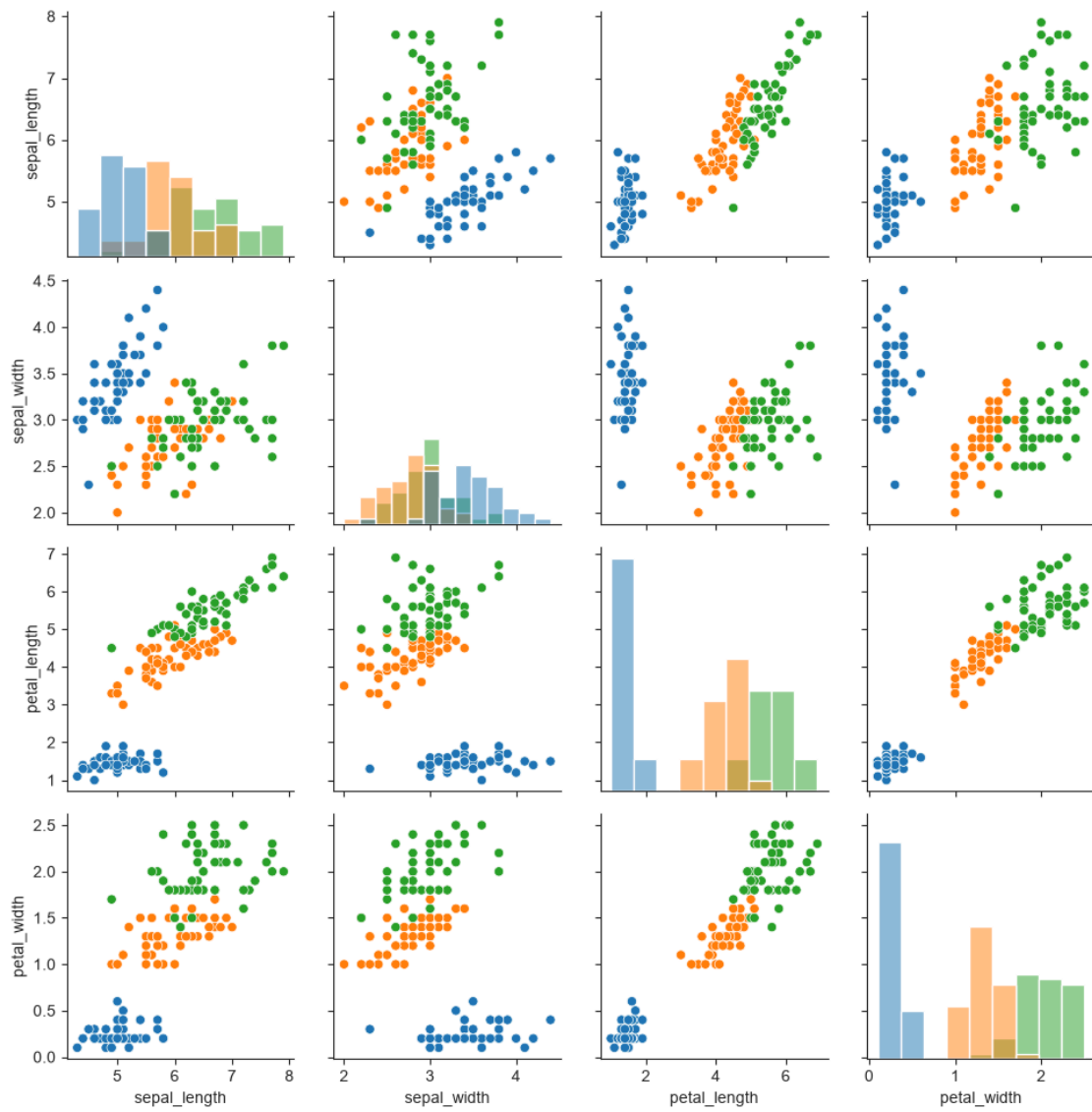


- Ticks Style + Muted:

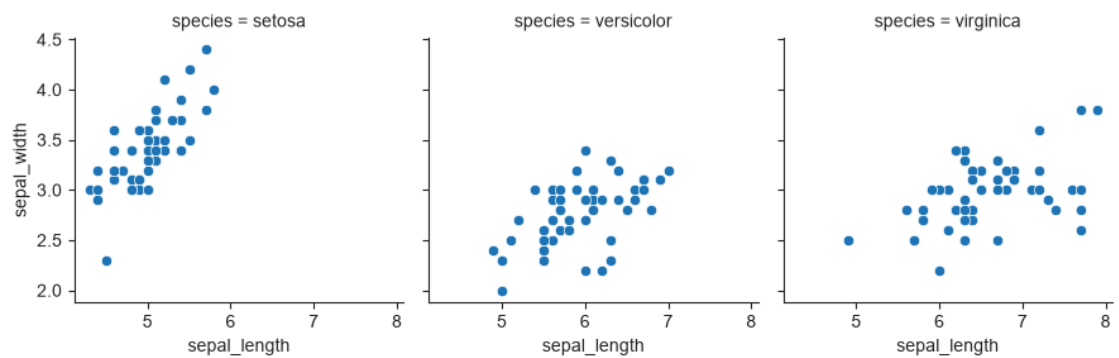


Visuals (Iris):

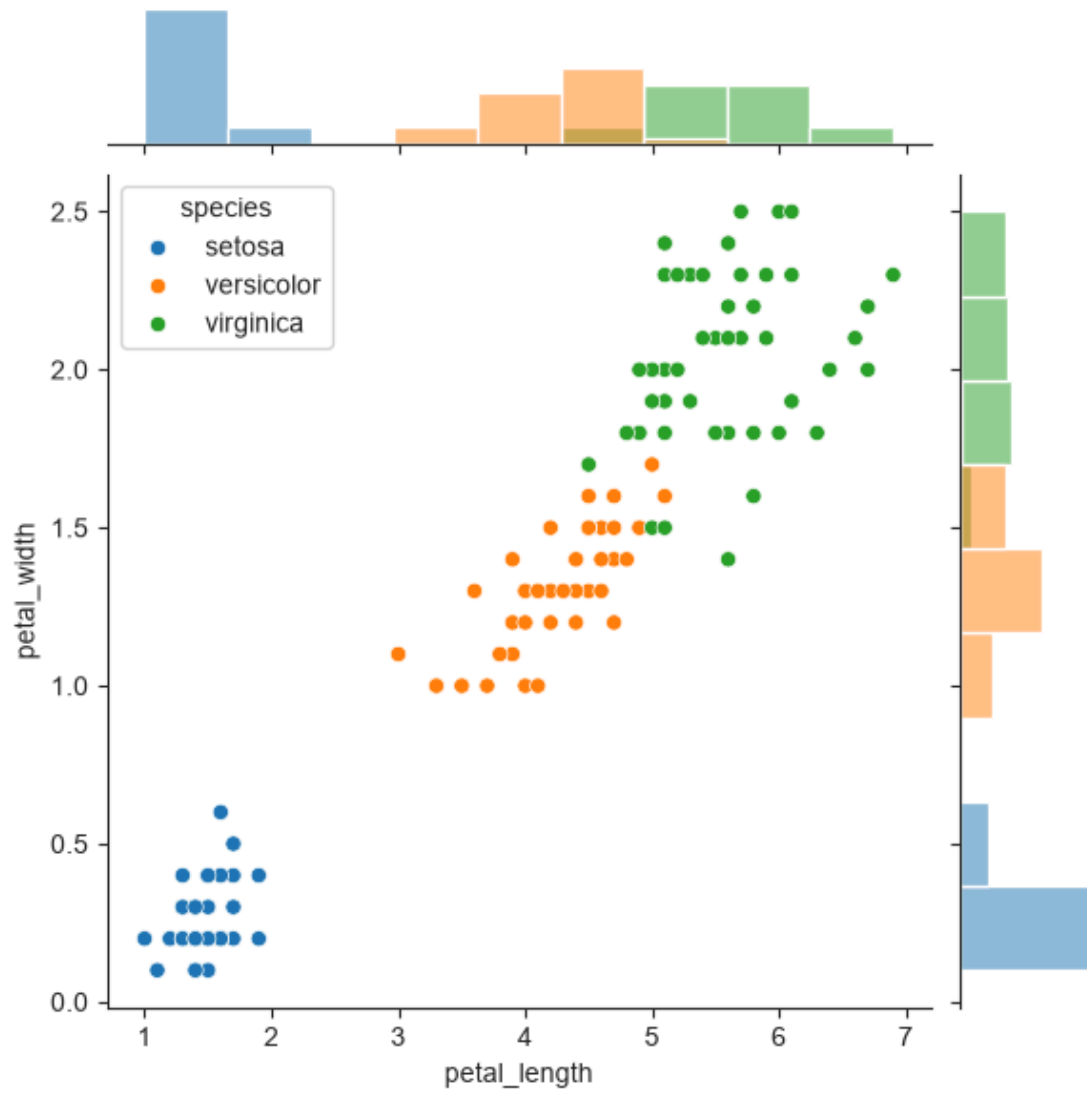
- PairGrid:



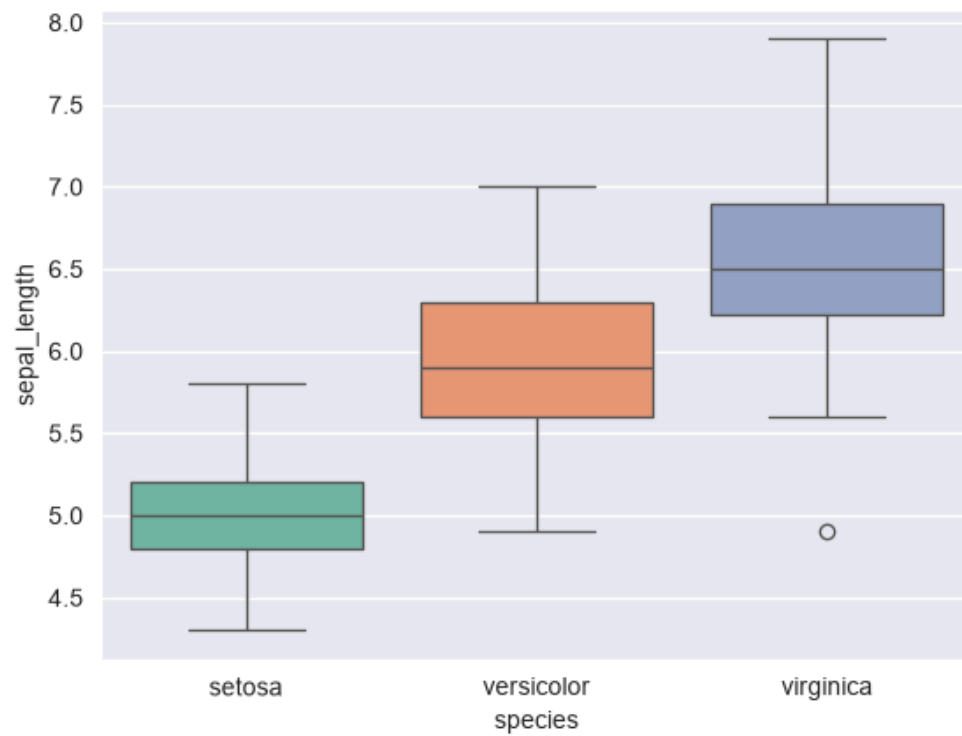
- FacetGrid:



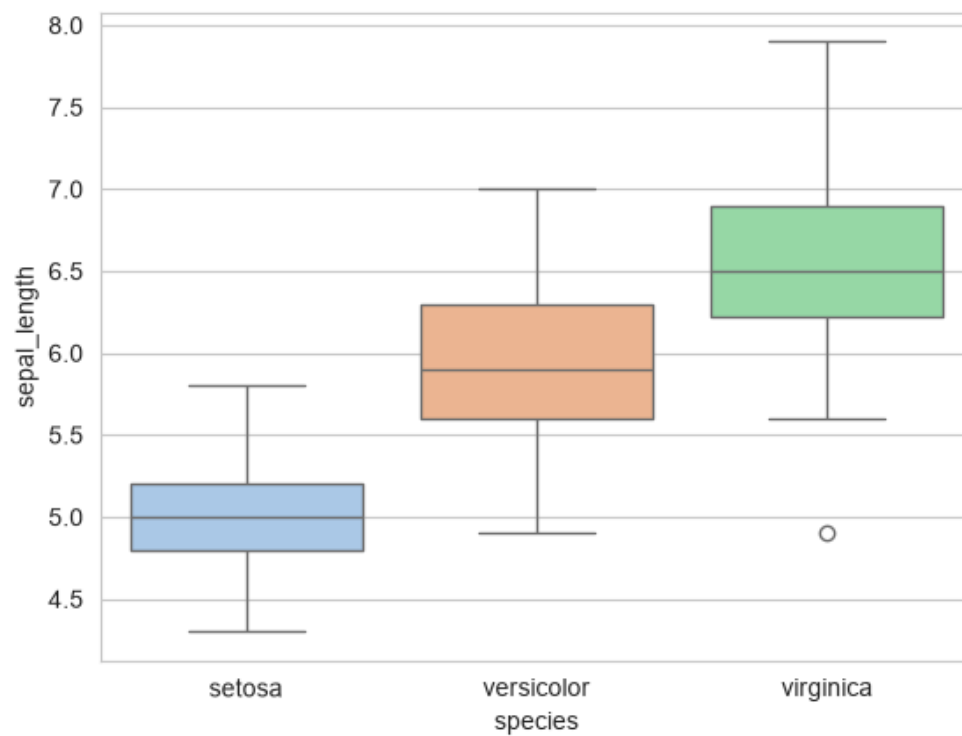
- JointGrid:



- Darkgrid Style + Viridis:



- Whitegrid Style + Plasma:



- Ticks Style + Muted:

